

CONTENTS

Scaffolding

04

4.1	Introduction	52
4.2	Important points	53
4.3	Planning and design	53
4.4	Basic independent scaffolding	56
4.5	Methods of access	64
4.6	Inspection and handover	65
4.7	Other considerations	67
Appendix A – Putlog scaffold inspection guidance		70
Appendix B – Independent tied scaffold inspection guidance		71
Appendix C – Birdcage scaffold inspection guidance		72
Appendix D – Case studies demonstrating the need for a good approach to scaffold management		73

Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

Overview

The purpose of this chapter is to give individuals an understanding of the safety issues that relate to the use of scaffolding, particularly site management with a responsibility for managing and controlling sites where scaffolding operations take place.

While the chapter provides relevant basic information, training in the actual techniques of erecting, altering or dismantling any scaffolding will be necessary before a person can be regarded as competent.

4.1 Introduction

This chapter will provide you with a basic knowledge of what to look for in the scaffold contractors you employ and the scaffolds that are erected. It will not confer competence in scaffolding techniques or scaffold inspection. Current industry training standards and requirements for different levels and types of scaffolding training are detailed within this chapter.

4.1.1 British Standard and technical guidance

BS EN 12811 (Part 1) *Temporary works equipment. Scaffolds – Performance requirements and general design* is a standard that specifies performance requirements and methods of structural and general design for access and working scaffolds.

Although this chapter applies to tube and fitting scaffolding, it is important to note that all proprietary scaffolding equipment (for example, system scaffolds, beams, modular transom units and hop-up brackets) must be manufactured and tested to the relevant British and European Standards (for example, BS EN 12810). All such equipment must also be used in accordance with manufacturer's instructions.

 **The British Standard can be accessed via the British Standards Institution website.**

Industry trade associations, including the Scaffolding Association (SA) and the National Access and Scaffolding Confederation (NASC), also publish advice, technical information and guidance for scaffolders and their clients on industry standards, scaffold design and health and safety in scaffolding operations. An example of this is the NASC publication *Good practice guidance for tube and fitting scaffolding* (TG20), which has been produced in response to the introduction of BS EN 12811 as a guide to good practice for all of those involved in the procurement, supply and use of tube and fitting scaffold. Although specifically for tube and fitting scaffolding, this guide can be used in the wider industry to increase awareness and make good practices accessible to all.

 **Visit the SA and NASC websites for further information.**

4.1.2 Training and competence

The term *competent person* is used in the Work at Height Regulations. However, although the regulations give no definition of competence, the Construction (Design and Management) Regulations 2015 (CDM) specify competence to be a combination of skills, knowledge, training and experience.

Competent person (for the purposes of scaffold inspection)

A **competent person** may be defined as a person who has both practical and theoretical knowledge, scaffold inspection training and actual experience of what they are to examine, in respect of a scaffold, so as to enable them to identify errors, defects, faults or weaknesses and to assess the importance of any such discovery.

A competent person must understand and recognise their limitations.

Competence (of individuals, for scaffold erection)

As regards the **competence** of individuals in relation to scaffold erection, competence may be taken to mean a person who has practical and theoretical knowledge, together with actual experience of scaffolding, and has acquired the recognised qualification in scaffolding.

This person must know their limitations and must not put themselves or others at risk.

The Work at Height Regulations require scaffolders to demonstrate specific training for the erection, alteration and dismantling of scaffolding.

Note: unless otherwise specified, all graphics are from TG20:21 with kind permission from the NASC.

4.1.3 Record schemes

4.1.3.1 Construction Industry Scaffolders Record Scheme (CISRS)

The CISRS record card scheme is managed and administered by CISRS Ltd. The scheme is the industry-recognised qualification for the scaffolding sector. All registered scaffolders are issued with an identification card that details the grade of their qualifications.

The card scheme categories include scaffolding labourer, trainee, scaffolder, advanced scaffolder, scaffolding supervisor and scaffold inspector (basic and advanced) qualifications. Scaffolding apprenticeships are also available for new entrant scaffolders. Options are available for tube and fitting scaffolding and system scaffolding. The system scaffold product training scheme (SSPTS) is for existing scaffolders who use specific proprietary system scaffolding products and can have their record card endorsed to demonstrate this. It is achieved by attending a two-day course on the type of system scaffold concerned.

The basic access systems erector (BASE) training is aimed at non-scaffolders who use system scaffold within a restricted scope of operation (for example, no public interface and a maximum height of 6 m). CISRS training makes it a requirement for any new entrant scaffolding trainee or scaffolding labourer to complete the CISRS construction operative training scheme (COTS) one-day course, before a card will be issued. Existing CISRS scaffolding labourer cardholders will be required to complete the COTS course upon expiry of their current card and prior to a new card being issued.



The CISRS website provides further information on UK-recognised training schemes within the scaffolding industry.

4.1.3.2 Other scaffolding training

Employers must demonstrate supervision by a competent person(s) and specific training for the use of proprietary products not covered by the CISRS scheme (such as temporary roofs, public access stairways, falsework and formwork systems).

4.2 Important points

- Scaffolding must only be erected, altered or dismantled by operatives who have been trained to do so or are under the direct supervision of a competent scaffolder.
- Employers who employ the services of scaffold contractors should look for proof of competence via the CISRS scheme.
- The main piece of legislation that applies to scaffolding is the Work at Height Regulations.
- Scaffolds are subject to statutory inspections, with reports of inspections raised where appropriate.
- All scaffolds must be properly erected, stable, tied to the structure as appropriate, suitable for their purpose and equipped with toe-boards and guard-rails on all working platforms.
- The BS EN 12811 and BS EN 13374 series of standards are supported by manufacturer's instructions, industry guidance (such as TG20) and bespoke design information for some structures.

4.3 Planning and design

4.3.1 Planning and preparation for a scaffolding contract

When selecting a scaffolding contractor you should check that they are competent, and consider the following.

- The competence of their management and operatives for the type of work.
- That they have a proven track record for the type of work.
- Their arrangements for managing health and safety (such as a recognised and auditable safety management system).
- Their past health and safety performance.
- That they have a positive safety culture.
- That they will allocate sufficient physical and human resources to service the contract (for example, scaffolding equipment, transport, qualified scaffolders, supervision and designers).
- Whether they are members of a recognised trade organisation (such as the SA or NASC).
- Whether they have any third-party health and safety or environmental accreditations (for example, OHSAS 18001/45001/14001 or accreditation through a Safety Schemes in Procurement (SSIP) members' scheme).

Before engaging a specialist scaffolding contractor, it is strongly advised that information is prepared on the intended use of the scaffolds for the contractor to take into account.



For further information on engaging and appointing specialist contractors and pre-construction information refer to Chapter A03 Construction (Design and Management) Regulations 2015.

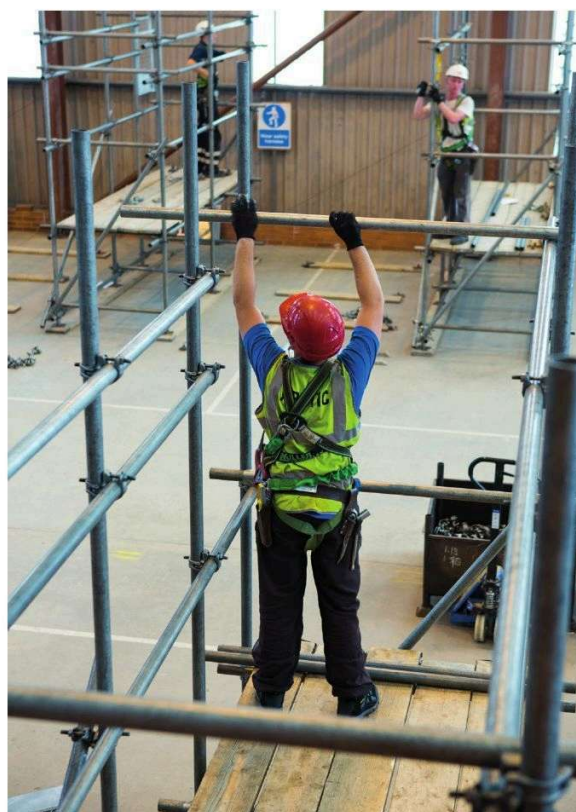
SCAFFOLDING

Before the erection of any scaffold, the following points should be clearly defined.

- What is the scaffold going to be used for and how long will it be required?
- What dimensions are required for the scaffold?
- Is the scaffolding compliant with BS EN 12811 and BS EN 13374? Is there a TG20 compliance sheet for the scaffold or is it met through manufacturer's instructions or bespoke design?
- Exactly where is it to be erected? Consider environmental factors (such as interface with the public or wildlife) and other areas (such as railway lines and overhead power lines).
- Can safe access be provided for the erection and use of the scaffold?
- How many working platforms will there be and will more than one working platform be loaded at any one time?
- What loadings will be imposed upon the working platforms, and on the scaffold as a whole?
- Are the ground conditions where the scaffold is to be erected suitable and capable of withstanding the intended load?
- How and where can the scaffold be stabilised (for example, ties)?
- Will the scaffold be clad with sheeting, netting or other materials (such as advertising hoarding)?

A pre-contract site inspection should be completed to understand the nature of the site and as the first part of the risk assessment process to identify any significant hazards.

This will allow suitable and sufficient control measures to be implemented and will help to ensure the safe and successful completion of the works.



Trainee scaffolders under instruction during training

4.3.2 Scaffold design

The Work at Height Regulations require strength and stability calculations for all scaffolds unless constructed to a generally recognised standard configuration.

TG20 provides guidance and definitions for compliant tube and fitting scaffolds, designed using structural calculation to BS EN 12811.

System scaffolding should be planned and constructed in accordance with manufacturer's instructions, to BS EN 12811 and BS EN 12810 along with BS EN 13374 *The standard for edge protection built with scaffold components*, otherwise the scaffold must be subject to bespoke design and calculation by a competent engineer.

Site management responsible for controlling scaffolding operations should seek assurance that the planned scaffolding conforms to a standard configuration or a bespoke design and that calculations have been prepared.

This can be achieved by the completion of a handover certificate, which will advise that the scaffold has been erected to the agreed specification, left in a condition suitable to perform the duty for which it is intended, and complies with all regulatory requirements.

The *Code of Practice for temporary works procedures and the permissible stress design of falsework* (BS 5975:2019) provides guidance on the procedural management of temporary structures, including scaffolding.

The TG20 design guide provides technical guidance for scaffolding design engineers on the application and interpretation of BS EN 12811 and related standards.

Scaffolding contractors and principal contractors should have suitable arrangements in place to manage and control the erection, alteration and commissioning of designed scaffolds (such as issuing drawings, managing variations to the design, inspection and handing over designed scaffolds).



For further information, including a list of scaffold structures where bespoke design is recommended, refer to the Health and Safety Executive (HSE) scaffold checklist.

4.3.3 TG20 compliant scaffolding

The TG20 operational guide contains a range of common scaffold configurations and scaffolding add-on features that do not require additional design. Any scaffold erected in accordance with TG20 is also in accordance with BS EN 12811 and BS EN 13374, and would not require a bespoke design.

A TG20 compliance sheet contains a brief summary of criteria that must be met in order for the scaffold to be considered compliant, and should be read alongside the operational guide. Types of TG20-compliant scaffolding used in the construction industry are shown below.

- Tied independent scaffolding.
- Freestanding independent scaffolding.
- Tied putlog scaffolding.
- Tower scaffolding, including lift shaft access.
- Interior access birdcage scaffolding.
- Cantilevered platforms and protection fans.
- Chimney-stack scaffolding.
- Loading bays.
- Ladder-access towers.
- Bridging with beams.
- Pavement lifts.
- Prefabricated transoms and hop-up brackets.

TG20 compliance sheets can be used for various purposes, including scaffold plans, risk assessments and method statements. They are not intended to replace the scaffolding handover certificate or permit to load; however, they can be appended to these documents issued by the scaffolding contractor.

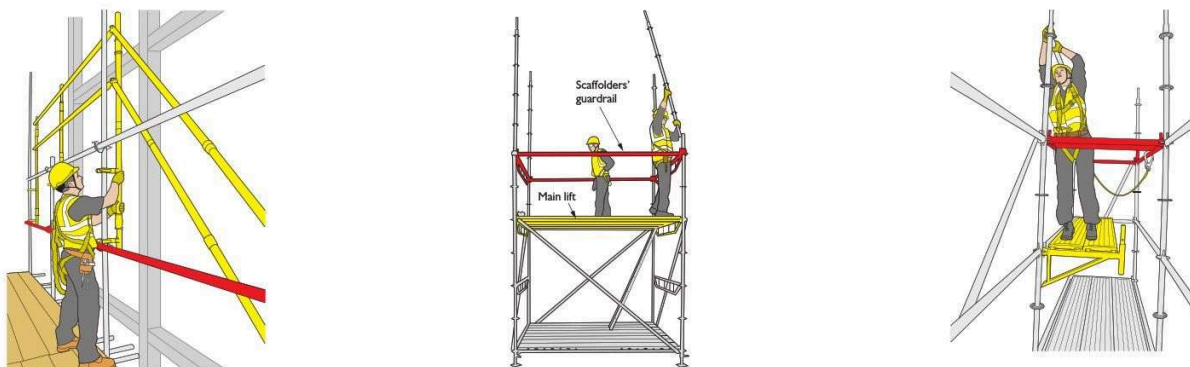
4.3.4 Fall prevention in scaffolding

Falling from height whilst erecting, altering or dismantling scaffolding presents a significant risk to all scaffolders. The HSE-endorsed NASC safety guidance *Preventing falls in scaffolding operations* (SG4), is the accepted industry standard to which all scaffolding contractors should be working. It highlights basic requirements and principles of fall prevention and protection that should be taken by scaffolders. The guide provides practical steps to follow to ensure compliance with the hierarchy of control measures specified by the Work at Height Regulations.

www.nasc.org.uk SG4 is free to download from the NASC website.

A range of collective fall protection methods and equipment are recommended in the SG4 guide (such as advanced guard-rail systems, step-ups and methods of working). These collective measures enable scaffolders to provide guard-rail protection in advance of erection and to maintain fall protection for alterations and dismantling. The use of temporary platforms with guard-rails should always take priority when establishing safe systems of work. SG4 refers to such temporary working platforms as the **scaffolders' safe zone**.

Such systems do not completely eliminate the risk of a fall in all circumstances and the guidance still recommends the use of personal fall protection equipment. SG4 recognises that personal fall protection equipment should be worn by scaffolders working at height and provides guidance for use as part of the systems of work, including the use of suitable anchor points.



Examples of collective protection equipment and methods and use of personal protection. Guard-rail shown in red is an integral part of the scaffolders' safe zone (Photo © NASC)

Personal fall protection equipment used for scaffolding should be inspected at the following intervals.

- Before use, by the user.
- Thoroughly by a competent person every three months and recorded.
- Thoroughly at other specified intervals if the need is identified via a risk assessment.



For further information on the management of fall protection equipment refer to Chapter D05 Fall arrest and suspension equipment.

SCAFFOLDING

4.3.5 Rescue planning

The Work at Height Regulations require employers to plan for emergencies and rescue for all work at height. These arrangements should not rely on the emergency services.

Scaffolding plans must also consider the rescue and recovery of a scaffolder suspended by their personal fall protection equipment.

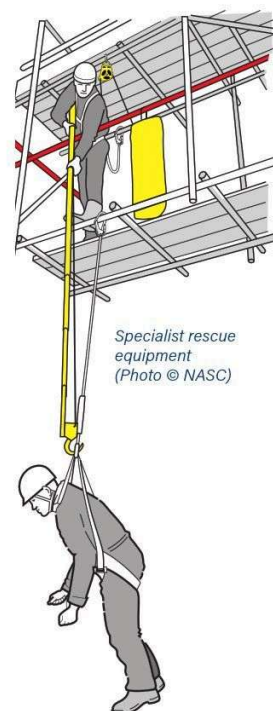
A rescue and recovery plan must be prepared as part of the risk assessment for each job, and the details must be included in the risk assessment and method statement shared with the client.

To ensure compliance with the above requirements:

- scaffolders and their supervisors should be trained in the requirements of PGN03 (latest version) and the most up-to-date version of the site specific work at height rescue plan, including the correct use of any specialist rescue equipment and techniques required
- site managers and others who run construction sites should check that the scaffold contractors coming on to their sites have been adequately trained in the requirements of PGN03 and SG19 (latest versions).

For further information on rescue planning refer to:

- [the Work at Height Safety Association's practical guidance note *Guidance on rescue during work at height* \(PGN03\)](#)
- [the NASC's safety guidance note *A guide to formulating a rescue plan* \(SG19\).](#)



4.4 Basic independent scaffolding

This section provides basic guidelines for independent scaffolding commonly used in the construction industry.

For detailed guidance refer to the manufacturer's instructions for proprietary scaffolding equipment or TG20 guidance.

4.4.1 Common materials and components

4.4.1.1 Tubes

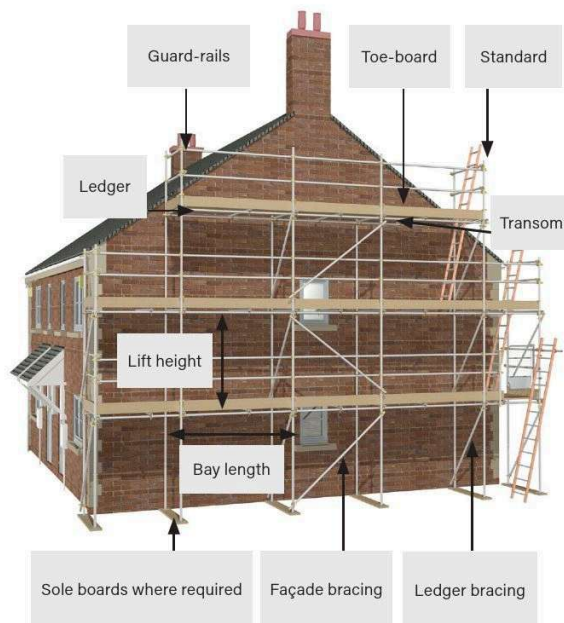
Tubes are made from cold formed high tensile steel and manufactured to British standard BS EN 39:2001, the technical standard for loose steel tubes for tube and coupler scaffolds.

Tubes come in two main types:

- Type 4 galvanised steel tubes or equivalent, 48.3 mm diameter with a 4 mm wall thickness.
- high tensile thinner wall 3.2 mm tubes made from hot or cold formed high tensile steel to BS EN 10210 (technical standard for hot finished structural hollow sections) or BS EN 10219 (technical standard for cold finished structural hollow sections). These standards set the minimum design load strengths for tubes.



An example of a galvanised steel tube (Photo © NASC)



Scaffolding terminology (Photo © NASC)

4.4.1.2 Fittings

- Couplers for tubes are manufactured to British standard BS EN 74-1:2005, the requirements and test procedures for couplers, spigot pins and baseplates for use in falsework and scaffolds.
- Old stocks of BS 1139 coupler can still be used. British standard BS 1139-2.2:2009(+A1:2015) provides the requirements and test methods for metal scaffolding, couplers and fittings (couplers and fittings outside the scope of BS EN 74).



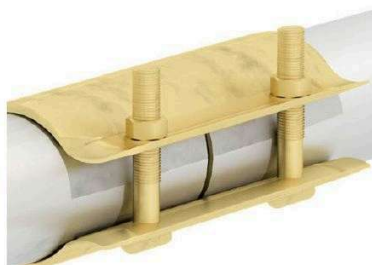
Right-angle coupler (double) (Photo © NASC)



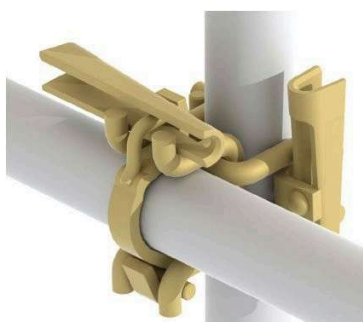
Swivel coupler (Photo © NASC)



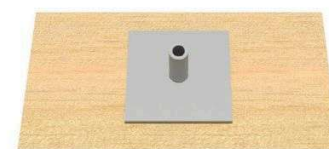
Putlog coupler (single) (Photo © NASC)



Sleeve coupler (Photo © NASC)



'Wedge' style right angle coupler (Photo © NASC)



Base plate (Photo © NASC)

4.4.1.3 Boards

- BS 2482 timber scaffold boards, 38 mm thickness, with a target span of at least 1.2 m.
- Laminated veneered lumber (LVL) or composite plastic decks may be used with a target span of 1.2 m and a unit weight not in excess of a 38 mm timber scaffold board.

Note: BS 2482 does not cover laminated or finger-jointed scaffold boards.



38 mm timber board (with end-bands) (Photo © NASC)



38 mm timber board (with nail-plate option) (Photo © NASC)

4.4.1.4 Ladders

- Ladders used with scaffolding should conform to BS EN 131 *Trade and Industrial – heavy duty and industrial use (for professional users)*.
- BS EN 131 replaces the British Standards for timber ladders (BS 1129) and aluminium ladders (BS 2037), which have been withdrawn. Ladders manufactured to these withdrawn standards may continue to be used, provided they are in good condition.
- Regulation 12 of the Work at Height Regulations requires work equipment to be inspected at regular intervals. When the ladder forms part of a scaffolding structure, the inspection should form part of the overall report for the scaffold.



External scaffold ladder access with safety gate (Photo © NASC)

4.4.1.5 Transom units



A structural transom unit (Photo © NASC)

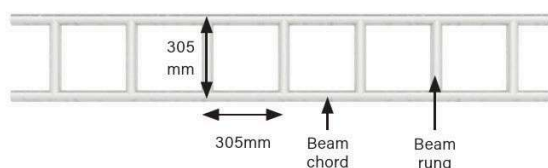


An extended intermediate transom unit (Photo © NASC)

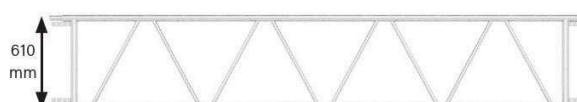
Prefabricated transoms are manufactured with integrated end connections fixed to both the standards and ledgers. As these connections are typically stronger than those of tube and putlog (single) couplers, in many cases scaffolding with these transoms can be erected without ledger bracing. They are available in a limited range of fixed lengths to accommodate the required number of scaffold boards and toe-boards between the standards.

Extending transom units are available in different width configurations. These components are adjustable, giving maximum on-site flexibility. They are designed for use as an intermediate board-bearing transom, which has an extendable section for inside board support. Many scaffolders value and benefit from the fast erection times, reduction in components and costs delivered through the use of these units.

A standard for prefabricated transom units has not been published. The NASC has defined and published a minimum specification for their structural performance, and manufacturers and suppliers can certify their transom units as TG20-compliant. If this is the case, they may be used to erect scaffolding in conjunction with a suitable TG20 compliance sheet.



Steel ladder beam (Photo © NASC)



Steel unit beam (Photo © NASC)

4.4.1.6 Beams

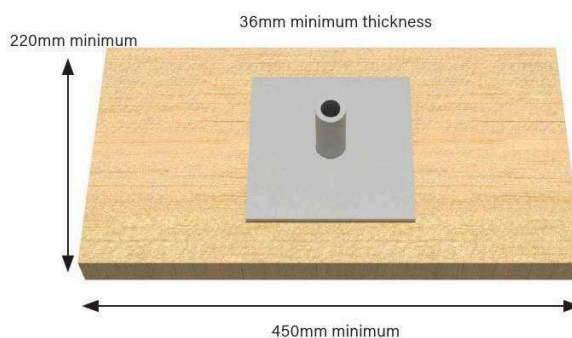
Scaffold beams are used with all types of scaffolding, to provide openings and bridges within a structure, to spread the load through the scaffold. A range of beam types is available, made from steel or aluminium (for example, ladder beams, lattice beams and X beams). Always check manufacturer's instructions when using beams with system scaffolding. Ensure the correct beams are used, as specified in bespoke designs or the TG20 compliance sheet provided for the work.

4.4.2 Foundations

The foundations for all scaffolds must be of adequate strength to support and disperse the applied load for the structure throughout its life. On hard surfaces (such as steel and concrete of sufficient strength and thickness) standards may be placed directly on the surface, although it is generally preferable to use a base plate that is 150 mm x 150 mm in size. Sheet piling or a proprietary plastic spreader plate can also be used to protect sensitive floors from damage or marking. They also serve to highlight the base plate to pedestrians. When supported by a typical pavement, a TG20-compliant scaffold with a maximum leg load of 12kN can be supported by base plates and plastic plates.



Typical high-visibility plate (Photo © NASC)



Minimum sole board dimensions (Photo © NASC)

On other surfaces, the load should be spread by using base plates and sole boards. The soil or ground beneath the sole board should be level and properly compacted. When a sole board is used on hard ground, the area beneath any one standard should be at least 0.1 m². If a timber sole board is used, it must be not less than 36 mm thick and a minimum dimension of 220 mm. Scaffold boards are typically used for this purpose. The scaffolding contractor should inform the client of the loads likely to be transmitted by the standards to the foundations, so that they may ensure that the ground or supporting structure is capable of supporting them.

The system scaffold instructions, bespoke design or TG20 compliance sheets should specify the requirements for scaffolding foundations. Planning should consider the loads likely to be imposed to any surface by scaffolding; this may include seeking specialist engineering advice. Planning should also include whether single or double sole boards are required and whether the scaffold requires double standards that may share sole boards.

For double standards a minimum sole board dimension of 750 x 225 x 75 mm is required, unless each standard has its own sole board.

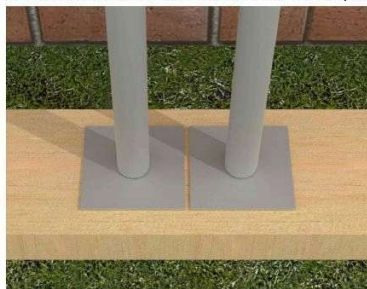


NASC has updated its TG20:21 guidance in line with current best practice on baseplates. The update confirms the position on baseplate thickness for scaffolding and access systems UK-wide, and is being made available to NASC members on its ePortal. It confirms the correct use of 2, 3 and 5 mm baseplates in various situations, and follows professional testing at an accredited test centre.

A foot lift (also referred to as a foot tie) is considered good practice and helps the stability of the scaffold, although it is not strictly necessary for most TG20-compliant scaffolds. This varies for system scaffolding, so refer to manufacturer instructions. TG20 compliance sheets may also specify that a foot lift is required for taller scaffolds. A foot lift is also recommended for sloping ground with a gradient of more than 1:10.



Double sole board (Photo © NASC)

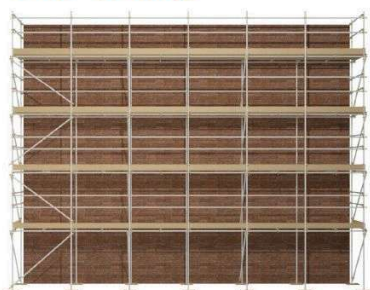


Double standards sharing a sole board (Photo © NASC)

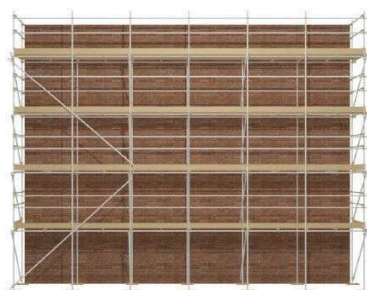


Foot lift example (Photo © NASC)

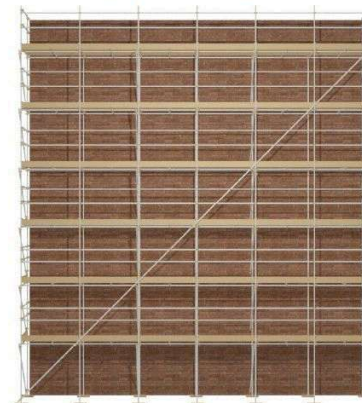
4.4.3 Bracing



Single-bay façade brace (Photo © NASC)



Double-bay façade brace (Photo © NASC)



Continuous façade brace (Photo © NASC)

For tube and fitting scaffolding, the TG20 operational guide specifies the requirements for bracing the various types of TG20-compliant scaffolds and scaffolding features (for example, pavement lifts, bridging with beams and loading bays). System scaffolding must be braced in accordance with the manufacturer's instructions. All other configurations must be braced in accordance with the bespoke design. The following examples are for basic tube and fitting tied independent scaffolds.

4.4.3.1 Façade bracing

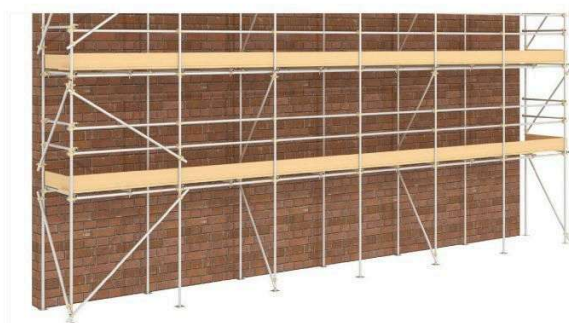
- Also referred to as sway bracing.
- Required between standards and ledgers on every elevation, generally every sixth bay horizontally. The six bay rule should be used as a guide because additional bracing is based on the number of bays, loading and sheeting requirements.
- Can be fixed as single-bay, double-bay or continuous.

4.4.3.2 Ledger bracing

- Connects inner and outer ledgers, or standards, to help stiffen and stabilise the scaffold. The first (end) pairs of standards and alternate pairs should be ledger braced at all lifts.

4.4.3.3 Plan bracing

- Not routinely required for basic independent scaffolds, except in two cases: bridges over openings supported with beams to help stabilise the beams; and pavement lifts, which provide clear access for pedestrians by omitting the bracing below the first lift.
- Other than these, it may be needed when erecting scaffolding that has been specially designed by an engineer, if the design requires it.



Ledger bracing (Photo © NASC)



Plan bracing at the underside of a lift (Photo © NASC)

SCAFFOLDING

4.4.4 Ties

A tie secures the scaffold to the supporting structure and is provided to resist inward (push) and outward (pull) movement, as well as to give some additional longitudinal stability (sway). Butting tubes may be required to provide sway resistance.

Ties are generally designated as moveable or non-moveable. Where possible, ties should be left undisturbed until the scaffold is dismantled. Where it is necessary for ties to be removed, even for a short period, the scaffold will be less stable and the fitting of additional temporary ties will be necessary, unless the initial tie-pattern was designed to allow for temporary removal.

Ties must not be removed by anyone other than a competent scaffolder or someone who is under the direct supervision of one. The removal of scaffold ties must be carried out in compliance with a safe system of work (such as a method statement and risk assessment). The manufacturer's information, scaffold design information (bespoke) or TG20 compliance sheets will provide details of the specification of ties to be used.

There are several methods of tying scaffolding to a building or other permanent structure, principally the assembly of tubes and fittings around structural features, or drilled or cast-in anchors to the building material. The capacity and load classes are stated in the TG20 operational guide. Refer to manufacturer's instructions for tying information of system scaffolding.

The specification, frequency and capacity must be determined from the TG20 compliance sheet for tube and fitting scaffolding, system scaffolding manufacturer's instruction manual or a design drawing. This will include the duty, frequency and maximum spacing of ties, which will vary due to the following factors.

- Location.
- Height.
- Load class.
- Lift height.
- Façade type.
- Cladding (brick guards, netting or sheeting).
- Seasonal variations.

At least 50% of ties should be fixed to ledger braced standards. Scaffolds erected around corners (returns) can provide stability and reduce the tying requirements. Refer to TG20 or the system manufacturer for details. Some typical methods are shown below.

4.4.4.1 Box ties

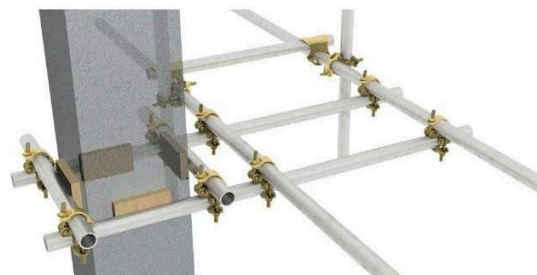
- Assemblies of tubes and fittings to form a square around a column or similar structural feature.
- Depending on its construction, it has variations that can be used for standard or heavy duty ties.

4.4.4.2 Through ties

- Fitted through windows or other façade openings.
- Suitable as a standard duty tie.
- Inner and outer tubes to resist movement and packing to protect the surfaces. If packing is used, it must be secured in position to prevent it falling and causing harm.

4.4.4.3 Lip ties

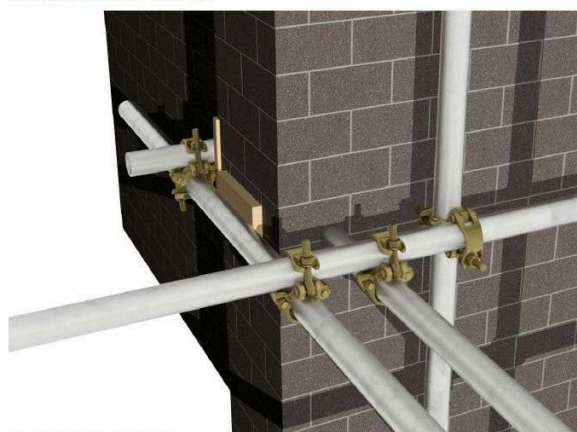
- Can be used if it is not possible to provide box or through ties.
- Suitable as a standard duty tie.



Box tie (Photo © NASC)



Through tie (Photo © NASC)



Lip tie (Photo © NASC)



Reveal tie (Photo © NASC)

- Opposing lip ties, butting tubes and/or sway transoms should be used to resist movement.

4.4.4.4 Reveal ties

- Used where other forms of tie are not practical.
- Only suitable as a light duty tie.
- A maximum of 50% of ties in a façade may be reveal ties.
- Reveal pins should be proof tested with a calibrated meter and regularly checked throughout the life of the scaffold to ensure they remain secure.

4.4.4.5 Anchoring methods

- Drilled or cast into the building fabric.
- Several types and methods of assembly (such as self-tapping screws and expanding plugs and sockets).
- Careful planning, selection, installation and testing are required for the correct use of masonry anchors (refer to the *Code of Practice for the selection and installation of post-installed anchors in concrete and masonry* (BS 8539:2012), *NASC guidance* (TG20 and TG4 Anchorage systems for scaffolding) and *manufacturer's instructions*).
- Scaffolders installing and testing anchors must be trained to do so.

4.4.4.6 Girder couplers

- Used to tie scaffolding to structural steelwork.
- They must be used in opposing pairs.
- The tie duty will depend upon their application and manufacturer's specifications.

4.4.5 Anchor testing

TG20 requires two levels of testing.

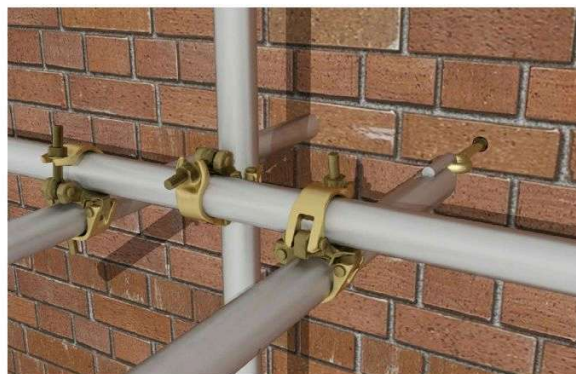
- Preliminary testing – to check the suitability of the fixing for the tie load required.
- Proof testing – to check the installation and that the required tensile loads can be achieved.

The preliminary tests should be carried out with a sample of five anchors for each base material within the project. If all test anchors support the load without slipping by 1 mm or more, the anchor may be used. For proof testing, a minimum of three ties must be tested and 5% (1 in 20) thereafter. If any anchor fails the test, the cause must be investigated and recorded. If more than two anchors fail then all of the ties (100%) must be tested. A test report should be provided with the test results as part of the handover process. The strength and pull-out capabilities of all cast or drilled anchorages must be confirmed before use.

4.4.6 Working platforms

The Work at Height Regulations and BS EN 12811, along with BS EN 13374 (the standard of temporary edge protection), set out minimum standards for working platforms for workers and users of scaffolding (Schedule 3 *Requirements for working platforms*). A higher standard is required for temporary scaffolding structures accessed by the general public and others (for example, temporary footbridge or fire escape) and specialist advice should be sought.

Working platforms should be assessed at planning stage to ensure there is sufficient space for access and working. BS 12811 stipulates a minimum width of 600 mm for work and 500 mm for access. TG20 recommends a minimum platform width of 800 mm for personnel and storage of materials.



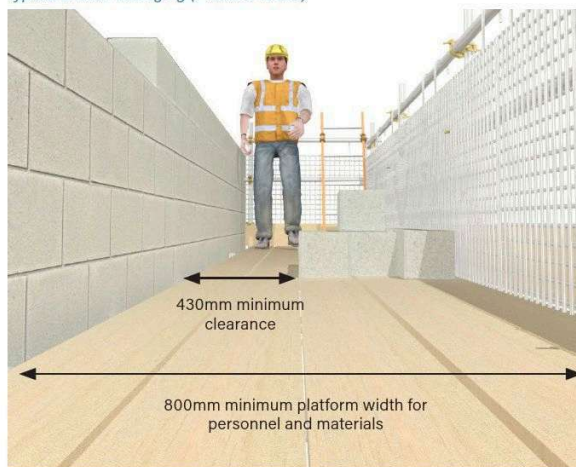
Smaller diameter ring bolts into masonry (Photo © NASC)



Girder clamps (Photo © NASC)



Typical anchor testing rig (Photo © NASC)



Suggested minimum clearance in TG20 (Photo © NASC)

SCAFFOLDING

Where materials are stored on a working platform TG20 also recommends that a minimum clearance of 430 mm be maintained for access.

Platforms are generally constructed using scaffold boards (BS 2482 or equivalent). Decking should be correctly supported, as recommended by TG20, the system scaffolding manufacturer's instructions or the design drawing. The spacing of transoms beneath scaffold boards is referred to as the target span and varies depending on the load class of the scaffold and specification of the board (for example, visually graded BS 2482 board load Classes 1-3 are 1.2 m and load Class 4 is 0.9 m).

The ends of boards must be supported to allow for deflection and to avoid accidental tipping (known as trap ends). Transoms must be positioned a minimum of 50 mm and no more than 150 mm from the end of the boards.

Note: 33 mm LVL boards have a maximum overhang of 130 mm.

All short boards less than 2.13 m (7 ft) must be secured at both ends. Rope, wire lashings or proprietary board clamps should be used. Overlapping of boards should be avoided, except at corner returns. Where an overlap is unavoidable, the lapped board should be secured to prevent movement and, if necessary, bevelled pieces should be installed (known as a fillet).

Gaps in platforms should be avoided, where possible, unless measures have been taken to prevent people from falling and falling objects from injuring someone. Where standards protrude through the platform (for example, between inside boards and the main platform) there will be a gap of approximately 50 mm. This gap is normally acceptable in construction. However, it may need to be covered following a risk assessment. There are several methods to cover such gaps. A competent scaffolding contractor would be able to advise on site specific requirements.

The space between the inner edge of a working platform and the adjacent structure should be kept as small as possible to prevent falls. However, there can be circumstances in which this gap has to be wider. This is usually due to the nature of the work being carried out (for example, to enable the craning-in of sections of curtain wall between the scaffold and the building under construction or where there is only primary steelwork inside the scaffold). This is known as the service gap.

In such circumstances, suitable protection measures must be taken if there is a risk of people falling, or people being struck by falling objects. A competent scaffolding contractor would be able to advise on site specific requirements. It is recommended that where a service gap exceeds 225 mm and a person could fall or suffer injury, double guard-rails and toe-boards should be installed or other suitable compensatory measures used (*see examples below*).

- Use of inner guard-rails and toe-boards.
- Segregation of the areas below the scaffold and post warning signs.
- Areas of the scaffold designated as danger areas, where access is restricted by guard-rails and warning signs.
- Use of personal fall protection equipment (work restraint or fall-arrest harnesses and suitable lanyards).

The space between scaffolding boards should be kept as small as possible and in any case should not exceed 25 mm.

Modern construction methods often require the use of inside boards to extend the width of the main platform. These inside platforms frequently need adapting to accommodate the construction work for various applications and trades (for example, they may require reducing to allow installation of the façade material). Inside boards can be supported by extending the main transoms, using telescopic transoms or using proprietary inside board brackets (known as hop-ups). Inside board brackets may be raised or lowered by up to 500 mm from the main platform. The inner board may need to be secured if there is a risk of accidental displacement.

Note: inside boards are generally rated as a lower load capacity to the main platform.

4.4.7 Guard-rails

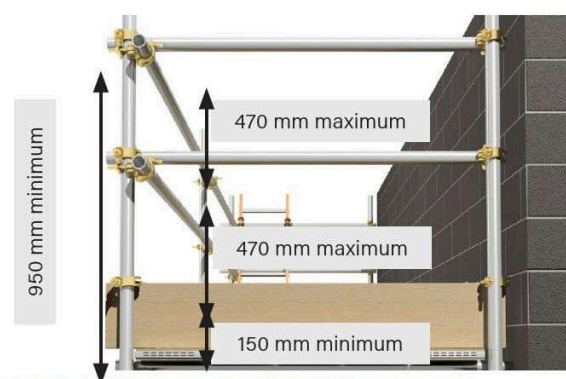
All scaffolding working platforms must be protected with double guard-rails where there is a risk of a fall. The main or top guard-rail must be 950 mm above the edge (or above a working platform) from which any person is liable to fall.

An intermediate guard-rail (or mid guard-rail) must be placed approximately halfway between the top edge of the toe-board and the top guard-rail, so there is no gap larger than 470 mm between the guard-rails or the toe-board.

There will be occasions when it is possible to fall from the working platform into the structure under construction. In these cases, it will be necessary to consider installing guard-rails to the inner edge of working platforms or using other fall prevention or arrest measures.

Additional guard-rails may be required to maintain the minimum height of 950 mm for a raised inside board bracket and other scenarios identified through a risk assessment. Where it is necessary to load out scaffolds with, for example, bulk materials, ideally there will be a purpose built loading bay with a lifting safety gate or similar. Where this is not the case, it is permissible to temporarily remove guard-rails and toe-boards in the following circumstances.

- Unless other work is stopped, other equally effective fall prevention or arrest measures are put in place (such as safety harnesses) whilst there is an exposed edge.
- Once the loading has been completed, the guard-rails and toe-boards are replaced immediately.



Work at Height Regulations guard-rail dimensions (Photo © NASC)

The removal and refitting of guard-rails and toe-boards must only be carried out by a competent scaffolder or someone who is under the direct supervision of a competent scaffolder.

! The removal and refitting of guard-rails and toe-boards must only be carried out by a competent scaffolder or someone who is under the direct supervision of a competent scaffolder.

4.4.8 Toe-boards

Toe-boards should be fitted to all boarded platforms where there is a risk of material or a person falling. The minimum height of a toe-board (as per BS EN 12811 and BS EN 13374) is 150 mm. In practice they are typically 225 mm scaffold boards, placed on edge and fixed inside the standard, at a minimum of two positions.

4.4.9 Brick guards

Brick guards should be installed where there is a possibility of materials toppling from working platforms and where materials will be stacked higher than the toe-boards. If using the common type of brick guard, intermediate guard-rails should also be fitted. The scaffold design or TG20 compliance sheets should take account of the use of brick guards and increased wind loading. Brick guards are now available in various materials and colours (for example, wire or plastic). Colour allows missing brick guards to be easily identified from the ground.

4.4.10 Loading

It is essential that scaffolds are not loaded beyond their maximum design load. Materials should be distributed as evenly as possible with heavy items (such as piles of bricks) positioned adjacent to standards. Overloading of working platforms can result in the failure of the boards or a scaffold collapse.

BS EN 12811 identifies six scaffold load classes, from 0.75 kN/m² to 6 kN/m². TG20 provides standard configurations for load Classes 1-4. Some system scaffolds may provide standard configurations for load Classes 5 and 6. A kilonewton (kN) is a unit of pressure, or force per area. 1 kN/m² = 100 kg/per square metre (approximately).

Regardless of how many lifts are boarded, only one lift may be fully loaded with personnel and material (100%) and one lift loaded to 50%. If it is necessary to simultaneously load more than two lifts, then a bespoke design is required.

TG20 states a load rating of 0.75 kN/m² for inside boards. Site managers should ensure that loading and use of scaffolding is properly controlled to prevent overloading. Where loading and storage of materials will exceed the load class, or it is to be loaded with mechanical handling equipment, then a specially strengthened section of the scaffold or a loading bay should be provided. Proprietary loading bays can be used in accordance with the manufacturer's instructions or TG20 compliance sheets for a compliant loading bay: otherwise, a bespoke design is required. The load rating and any special restrictions should be clearly displayed on the loading bay.

4.4.11 System scaffolding

- Manufactured to BS EN 12811 and BS EN 12810.
- There are various types of system scaffolding available and these must be used in accordance with the manufacturer's instructions.



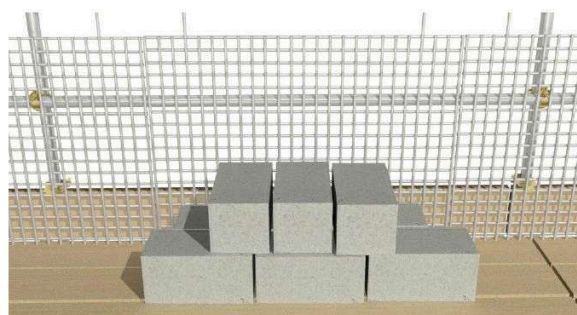
Pressing type (such as Kwikstage)



Cup type (such as Cuplok)



Rosette type (such as Layher)



Wire brick guard with double guard-rails and toe-board (Photo © NASC)



TG20 loading bay (Photo © NASC)

SCAFFOLDING

All scaffolding material must be supplied in good condition and inspected before use. Scaffolding contractors should have systems in place to ensure scaffolding equipment is routinely inspected, serviced and, if necessary, repaired by a competent person.

4.4.12 Freestanding access scaffolds

Some scaffolds are designed to be used as freestanding structures, such as those listed below.

- Stationary freestanding towers.
- Chimney stack access scaffolding.
- Freestanding independent scaffolding (supported by rakers).
- Interior access birdcage scaffolding.

Freestanding access scaffolds or sections of scaffolds for modern methods of construction (such as timber-frame type structures) or demolition require calculations to be made for stability. These scaffolds would therefore be classed as special scaffolds and require bespoke design and calculations. Engineers can utilise alternative methods of achieving stability (such as bracing, kentledge (ballast or counterweights), guys and ground anchors). Returns around corners, access towers and loading bays can also be taken into account by engineers when calculating stability.



Rakers connected at the outer face for a freestanding scaffold (Photo © NASC)

4.5 Methods of access

The **Work at Height Regulations** require employers to specify the use of existing structures as a means of access to height (for example, lifts or a permanent staircase) in preference to temporary measures (such as ladders). Whilst ladders have been the commonly used means of access to scaffolds, the use of other, safer means of access (such as stair towers) should now be considered in preference. This would also aid timely evacuation in the event of an emergency (such as a fire).

BS EN 12811 recommends that where extensive work is carried out, stairways should be provided for access, and for taller scaffolds consideration should be given to the use of a passenger hoist.

Where passenger hoists are used, additional non-mechanical access must also be provided in case of any breakdown or emergencies. The following hierarchy of access is recommended for all scaffolds.

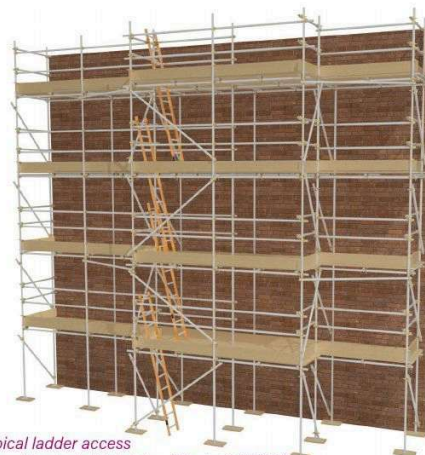
1. Staircases or lifts in permanent structures.
2. Mechanical hoists (for larger structures with additional emergency access).
3. Temporary stairways.
4. Ladder access towers with single lift ladders (to reduce the potential fall distance).
5. Ladder access bays with multiple lift ladders.
6. Internal ladder access with protected ladder traps.
7. External ladder using a safety gate.

Straight ladders used for access to a scaffold must meet the following requirements.

- Be manufactured to BS EN 131, BS1129 and BS2037 (see Section 4.4.14 for more detail).
- Not be defective in any way.
- Not be painted or treated in any way that might hide defects.
- Be placed on a firm footing, with each stile equally supported.
- Be so positioned that there is sufficient space at each rung to give an adequate foothold.



A typical site stairway (Photo © NASC)



A typical ladder access tower with single lift ladders (Photo © NASC)

- Be positioned approximately at an angle of 75°; that is, one measure horizontal to four measures vertical.
- Be secured at the top using square lashings or a proprietary ladder clamp. For longer ladders, additional ladder supports (stays) can be used to prevent the ladder deflecting when used. The stay must not obstruct the rung of the ladder.
- Extend approximately 1 m above the working platform, unless there is another adequate handhold.
- Be provided with suitable rest platforms if rising more than 9 m.
- Be the right way up (tie wires or bars positioned under the rungs).

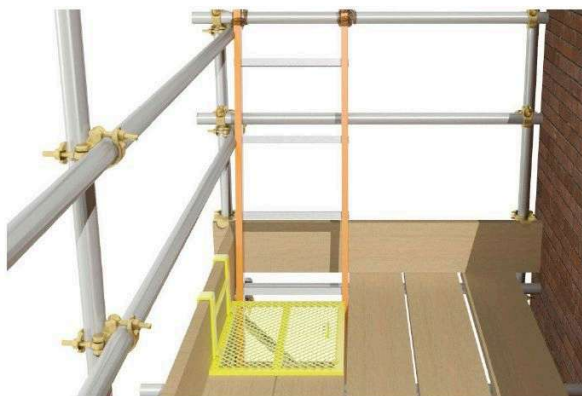
Shorter ladders, which provide access to only one lift, are recommended to reduce the potential fall risk, although the Work at Height Regulations allow a maximum vertical height of 9 m between rest platforms.

Where scaffolds are designed with internal ladders, working platforms must be provided with access holes for each ladder.

Such access holes should be at least 450 mm wide (across the platform) and not less than 600 mm in the other direction.



External access ladder and safety gate (Photo © NASC)



Typical ladder trapdoor (Photo © NASC)



An alternative ladder trap protection (Photo © NASC)



The access hole should be protected with a ladder trapdoor or similar (such as guard-rails and a gate).

Lateral gaps in guard-rails and toe-boards for access and egress must be kept to a minimum and protected with a ladder safety gate.

4.6 Inspection and handover

All scaffolds and working platforms (together with all other protective measures) are work equipment for the purposes of the Work at Height Regulations and so the inspection requirements of the regulations apply, which state:



... every employer shall ensure that where the safety of work equipment depends on how it is installed or assembled, it is not used after installation or assembly in any position unless it has been inspected in that position.

Employers must ensure that working platforms used for construction and from which a person could fall more than 2 m are not used in any position unless they have been inspected in that position within the previous seven days. Additionally all scaffolding, working platforms and edge protection should be inspected before being used for the first time and following significant alteration

All scaffolding inspections should be carried out by a competent person whose combination of knowledge, training and experience is appropriate for the type and complexity of the scaffold. Competence may have been assessed under CISRS or an individual may have received training in inspecting a specific type of system scaffold from a manufacturer or supplier.

An employer may wish to have a commercial arrangement with a scaffolding contractor or independent consultant in order to carry out inspections.



For further information about scaffold inspection qualifications refer to the CISRS scaffold inspection training scheme (SITS).

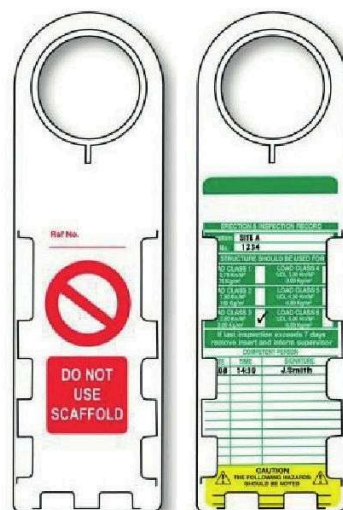
SCAFFOLDING

There is a further requirement that every employer must ensure that work equipment exposed to conditions that may cause any deterioration liable to result in dangerous situations is inspected at suitable intervals, and also when any exceptional circumstances that are liable to jeopardise the safety of the work equipment have occurred (such as severe weather conditions or vehicle impact).

It is common for proprietary inspection tags to be used to provide a status report for the scaffold. Where tags are used it is important that one is positioned at every access point to the scaffold.

4.6.1 Inspection of places of work at height

An additional requirement as regards inspection (or checking) is that every employer must ensure, so far as it is reasonably practicable, that every surface, parapet, permanent rail or other such fall protection measure of every place of work at height are checked before each use. While there appears to be no requirement to record these checks, the prudent employer should do so to prove (for example, to the HSE or clients) that they have in fact been done.



Typical scaffold tagging system

4.6.2 Reports

Where an inspection has been carried out, a written report of the findings must be made before the end of the working period. The person who prepares the written report must provide a copy within 24 hours to the person on whose behalf the inspection was carried out. The report, or a copy of it, must be kept on the site where the inspection was carried out until the construction work is completed, and then kept at the company offices for three months. Employers are free to design their own inspection report forms or purchase pads of them from commercial suppliers. In either case, Schedule 7 of the Work at Height Regulations says they must include the following information.

- The name and address of the person for whom the inspection was carried out.
- Details of any matter identified that could risk the health or safety of any person.
- The location of the work equipment inspected.
- A description of the work equipment inspected.
- The date and time of the inspection.
- Details of any action taken as a result of any matter identified above.
- Details of any further actions considered necessary.
- The name and position of the person making the report.

When an inspection has been carried out, and it has been identified that the structure has been interfered with or is not safe, the first priority is to ensure the safety of anyone who may be working on, near to or intending to work on the structure. Everybody must be taken off the structure, by a safe means, and access to it must be restricted until it can be inspected again - and, if required, components replaced or defects rectified by competent operatives.

4.6.3 Handover

After completion or following a modification to the scaffolding, a competent person must carry out an inspection and complete a report. It is good industry practice for scaffolding contractors to then complete a scaffold handover certificate and ensure that the client's representative receives a copy. Where a tagging system is used a green or red tag must be completed after each inspection and inserted into a holder at every access point of the scaffold. Some sites may require the report completed in the form of a site register.

Scaffolding contractors should include supporting documents (such as those listed below) with the handover certificate.

- TG20 compliance sheet(s).
- Bespoke design drawings and calculations.
- Anchor testing reports.

4.6.4 Incomplete scaffolding and danger areas

Where the erection of a scaffold has not been completed, physical measures must be taken to restrict access (for example, remove or board over the access ladder(s) and warn people of the fact that the scaffold is not safe to use). The method of warning is usually a 'Scaffold incomplete' sign at each point of access.

Warning and prohibition signs must conform to the requirements of the Health and Safety (Safety Signs and Signals) Regulations 1996. Any scaffold structure, or part of (for example, during alterations to a scaffold when part of the structure remains in use), and a workplace where scaffolding operations are being undertaken, where there exists a risk of a fall



Good example of a warning incorporated into a physical barrier (Photo © NASC)

or a risk of a person being struck by a falling object, shall be defined as a danger area. Access to danger areas should be restricted and communicated effectively to all concerned (for example, by guard-rails or other suitable barriers and signs).

www For further information refer to *Access and egress from scaffolds (SG25)*.

4.7 Other considerations

4.7.1 Hoists with scaffolding

Special goods and personnel hoists are often used with scaffolding. Ideally mechanical hoists should be fixed to the permanent structure unless the scaffold has been specifically designed for the purpose. NASC and the Construction Plant Association (CPA) have produced specific guidance for the use of hoists for transporting scaffolding materials during erection, alterations and dismantling (*for further information refer to SG26*). Mechanical and non-mechanical winches are commonly used both for scaffolding operations and by the users of scaffolding (for example, gin wheels and ropes).



A selection of typical hoists and winches used with scaffolding

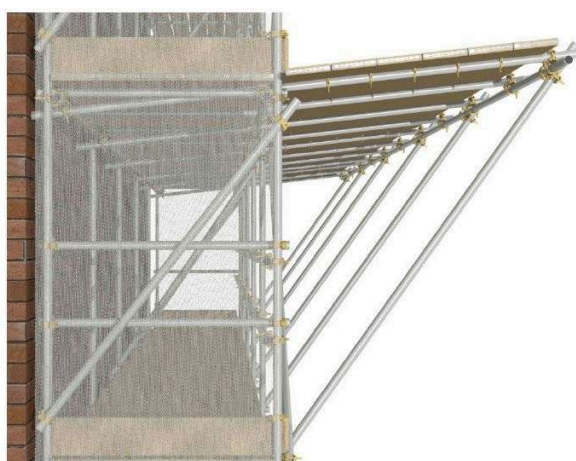
4.7.2 Protection of the public

Protection of the public and other persons against falling materials should be provided by the use of nets, brick guards, toe-boards, protective fans, and so on. The precautions to be adopted should be identified at the planning stage of the scaffold provision, including the risk assessment process.

Guidance on protection of the public (SG34) provides general advice for scaffolds erected in the public domain, including the planning required and the range of precautions that should be considered to eliminate the likelihood of harm (for example, through falling material or transport accidents).

www TG20 includes compliant scaffolding features (such as pavement lifts up to 2.7 m and light-duty protection fans).

Reference should also be made to the HSE guidance document *Protecting the public: Your next move (HSG151)*.



A TG20-compliant protection fan (Photo © NASC)



TG20 pavement lift (Photo © NASC)

Local Authority permission will normally be required for scaffolds to be built on the pavement or footpaths. Scaffolds can cause particular problems to people with physical and visual disabilities, unless adequate steps are taken to reduce contact hazards with items such as tube ends and threads on fittings.

SCAFFOLDING

The needs of children, the elderly and the disabled should be considered: for example, ramps for wheelchairs and tapping boards may be beneficial to warn visually-impaired people of the hazards.

This can usually be achieved by cladding the run of standards or binding the scaffold poles and fittings with suitable padding and/or brightly coloured bunting and tape (black and yellow are recommended for people who are sight impaired).

All joints should be wrapped to protect the public and other persons from sharp edges, and plastic protective cups should be placed in or over the ends of tubes. Both of these precautions should be carried out to a height of 2 m above the ground level, and Local Authority minimum clearances for the lift dimensions must be met.

It is not sufficient for employers to warn the public of the hazard just by placing safety signs on the scaffold, even if they meet the requirements of the Health and Safety (Safety Signs and Signals) Regulations. Employers must have acted to reduce the hazard as far as reasonably practicable.

To ensure the scaffold is not struck by vehicles, no part of it should be allowed to project into the roadway, unless appropriate measures are taken, such as traffic control or a road closure. The provision of lighting at night may also be necessary, as specified by the Local Authority.

Supplementary lights should be installed on the scaffold where it has been erected in a place to which the public has access (pedestrians or vehicles), unless the level of background lighting after dark is thought to be sufficient. Pavement gantries will require bespoke design and calculations.

4.7.3 Sheeting and netting

Sheeting, netting or hoarding should not be attached to scaffolding, unless the scaffold has been designed and calculated to resist the additional wind loading. Some proprietary scaffold sheeting includes sacrificial ties designed to break in tension at a certain load.

Planning for clad scaffolding needs to consider fire risk from sheeted scaffolding, particularly on occupied premises. Flame-resistant sheeting or netting should be specified and conform to the Loss Prevention Certification Board (LPCB) standard LPS1215, BS 5867-2, BS EN 13501 or equivalent.

4.7.4 Rubbish chutes

Rubbish chutes (also referred to as debris chutes) may be fixed to a scaffold. However, the scaffolding supporting the chute assembly is a non-standard scaffolding, and it therefore requires a design check for strength and stability. The chutes should be installed and inspected in accordance with the manufacturer's instructions.

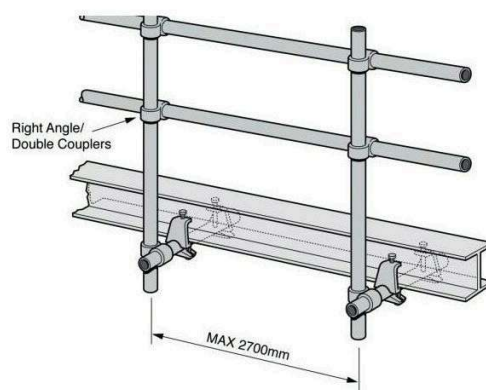
Site managers should ensure that chutes are regularly checked for blockages as the increased load could affect the stability of the chute and the scaffolding.

4.7.5 Temporary edge protection (scaffolding)

Temporary edge protection constructed from scaffolding should comply with *Temporary edge protection systems* (BS EN 13374).

For further information refer to the HSE guidance *Health and safety in roof work* (HSG33), the Edge Protection Federation's *Code of Practice 2014 – A guide to the selection and use of temporary edge protection systems*, and NASC's *TG1:22 Temporary edge protection*.

www



Example of temporary scaffolding edge protection

4.7.6 Unauthorised modifications to scaffolds

4.7.6.1 Maintaining safe scaffold structures

The consequences of any unauthorised alterations or modifications to scaffolds could result in a fatality or serious injury to contractors, the general public or yourself and may result in damage to adjacent property. The modification of scaffolds by non-qualified scaffolding operatives (a practice also known as *interference*) is unacceptable under any circumstances and may lead to prosecution by the enforcing authorities.

Scaffolds should only be modified by competent scaffolders who have been authorised to do so by the scaffold contractor.

It is also unacceptable for the client and/or user to authorise alterations without prior consent from the scaffold contractor, as it may invalidate their insurance cover and could be an offence under Sections 7 and 8 of the Health and Safety at Work etc. Act 1974.

Most contractors and principal contractors now enforce a zero tolerance policy in regards to unauthorised scaffold interference by their contractors, and work closely with the scaffolding contractor to ensure it is applied.

The most common types of scaffold interference are the removal of scaffold structural ties by other trades, the removal of handrails and toe-boards to allow materials to be loaded directly onto the working platforms and the undermining of the scaffold foundations by utility contractors. Good planning and communication with all contractors will help prevent unauthorised scaffold modifications.

The guidance below will help the user in assuring that scaffold structures are, and remain, fit for purpose.

- All scaffolds shall be erected in accordance with statutory requirements and in accordance with the manufacturer's instructions when using system scaffolds.
- Employers must provide appropriate levels of supervision, taking into account the complexity of the work and the levels of training and competence of the scaffolders involved.
- Employees should be competent for the type of scaffolding work they are undertaking and should have received appropriate training relevant to the type and complexity of scaffolding they are working on.
- As a minimum requirement, every scaffold gang should contain a competent scaffolder who has received training for the type and complexity of the scaffold to be erected, altered or dismantled.
- Erection, alteration and dismantling of all scaffolding structures (basic or complex) should be carried out under the direct supervision of a competent person.
- All mobile tower scaffolds shall be erected by a competent person who is in the possession of a Prefabricated Access Suppliers' and Manufacturers' Association (PASMA) qualification or other recognised qualifications.
- All structures must be handed over, by a competent person, to the customer. In some cases a tag-type system is used at the ladder access points to clearly show the validity and suitability of the structure. Scaffolding should be inspected:
 - before it is put into use
 - at seven-day intervals until it is dismantled
 - after bad or excessively wet weather or high winds or another event likely to have affected its strength or stability
 - after any substantial additions or other alterations
 - after unauthorised interference.
- A written report must be prepared by the competent person. A copy of the report should be kept on site and a further copy be retained for a period of three months from the completion of the work with the person on whose behalf the inspection was carried out. If a scaffold fails inspection this must be reported by the person carrying out the inspection, to the person responsible for the scaffolding, as soon as possible.

4.7.6.2 How to identify unauthorised modifications

The following information is not an exhaustive list but should help you to recognise good practice before and whilst using scaffold structures.

- Check that the foundations have not been disturbed or undermined and the standards are on base plates and sole boards (as necessary).
- Check that guard-rails are not missing and they are installed on every lift in all areas in which falls could occur.
- Check that toe-boards have not been removed or displaced.
- Check that any scaffold boards have not been removed, displaced or damaged (such as disc cutter marks).
- Check that transoms are suitably placed to support the boards and that the maximum support span identified on the board end plate has not been exceeded.
- Check that scaffold ties have not been removed.
- Check that any bracing is not missing.
- Check that ladders are properly secured.
- Check that any brickguards and/or netting, where fitted, are still in place.
- Check that the structure is not being overloaded. The safe load should be clearly marked on the scaffold tag.
- Check for any other signs of unauthorised interference. If found, report them to the scaffold contractor immediately.

4.7.6.3 Action to be taken if concerned

If for any reason you identify that the structure has been interfered with, the first priority will be to ensure the safety of any contractors who may be working, or intending to work, on the structure.

All users must vacate the structure by a safe means and access to the scaffold should be restricted until such time that it has been inspected and, if required, the components replaced by competent scaffolding operatives.

Appendix A – Putlog scaffold inspection guidance



Safety checklist

Check from the ground (see numbers above)

1. Base soundness; adequate spread of load; particularly as there is only one line of standards, avoidance of pavement lights, inspection chambers, and so on; no nearby excavations.
2. Line of standards and ledgers. Standards vertical.
3. Spacing of putlogs.
4. Working platform. Check line and even support of boards; overhang; lapped boards and fillets.
5. Guard-rails and toe-boards.
6. Security of boards, toe-boards and guard-rails.
7. Longitudinal bracing.
8. Means of access.

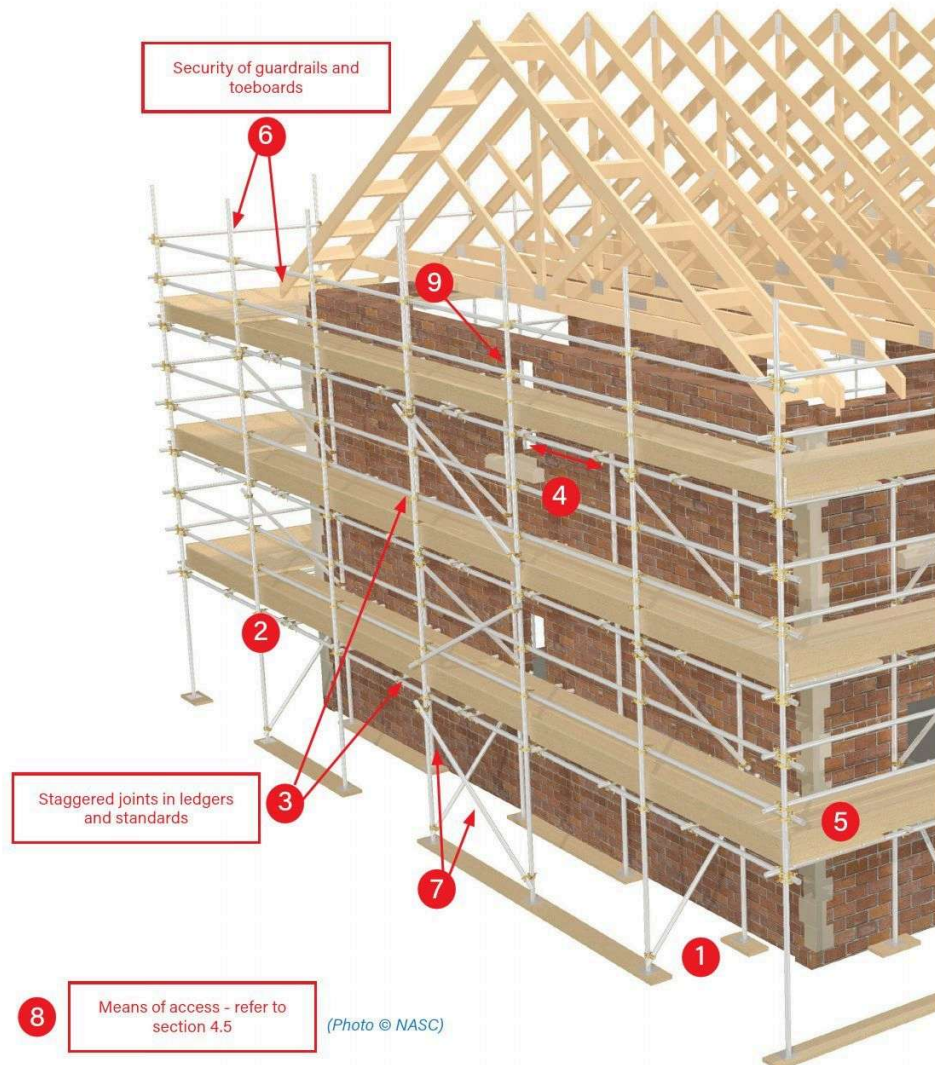
Check from inside building or on the scaffold

9. Spade end of putlog laid horizontally where possible, fully home (75 mm) in brickwork (bed joint).
10. Ties, particularly on lift below working platform or, in early stages, rakers on alternate standards. Special attention to 'through' ties on large flank ends without windows. Load-bearing couplers to be used.
11. Platform loading (not overloaded).
12. Security and correct use of all fittings (couplers), particularly on transoms and bracing.
13. Condition of tubes and fittings.
14. Damage from falling material.
15. Security of boards, toe-boards and guard-rails.
16. Security of stacked materials.

Internal edge protection has been omitted for clarity.

Refer to Section 6b of TG20 for additional information on putlog scaffolds.

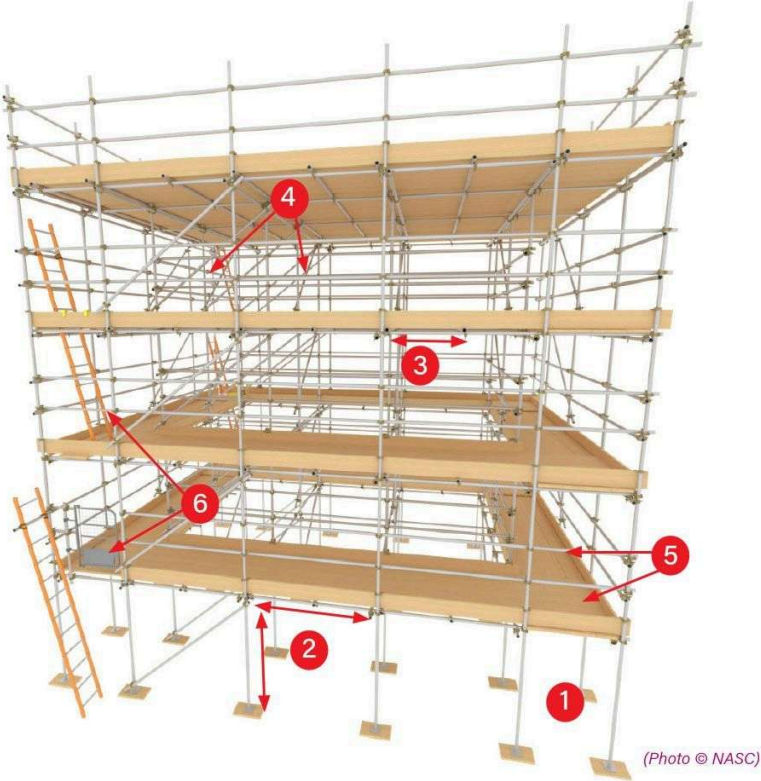
Appendix B – Independent tied scaffold inspection guidance



Safety checklist	
Check from the ground (see numbers above)	Check from the scaffold
1. Base soundness; adequate spread of load; avoidance of pavement lights, inspection chambers, and so on; no nearby excavations.	9. Ties, particularly on lift below working platform or, in early stages, rakers on alternate standards. Special attention to 'through' ties on large flank ends without windows. Load-bearing couplers to be used.
2. Line of standards and ledgers; standards vertical.	10. Special loadings by protective fans, wind sails, and similar; anchorage and spread of load.
3. Staggering of joints (vertical and horizontal).	11. Security of boards, toe-boards and guard-rails.
4. Spacing of transoms.	12. Security and correct use of all fittings (couplers), particularly on transoms and bracing.
5. Working platform. Check line and even support of boards; overhang; lapped boards and fillets.	13. Condition of tubes and fittings.
6. Security of guard-rails and toe-boards.	14. Damage by loads swinging from cranes or by falling material.
7. Longitudinal, ledger and plan bracing.	15. Overloading.
8. Means of access.	16. Security of stacked materials.
Refer to Section 6 of TG20 for additional information on independent tied scaffolds.	

Appendix C - Birdcage scaffold inspection guidance

04



Safety checklist (1-6 above, 7-11 in section 4.4)	
1. Base.	7. Security and correct use of couplers and fittings.
2. Line of standards and ledgers.	8. Condition of tube and fittings.
3. Line and spacing of transoms.	9. Even spread of load on platform.
4. Diagonal bracing (in both directions).	10. Overloading.
5. Security of boards, toe-boards and guard-rails.	11. Security of stacked materials.
6. Means of access.	
Refer to Section 13 of TG20 for additional information on interior birdcage scaffolds.	

Appendix D – Case studies demonstrating the need for a good approach to scaffold management



Independent tied perimeter scaffold collapse

An independent tied perimeter scaffold collapsed at a construction site in Milton Keynes. The collapse started on the west elevation, with a partial collapse of the north elevation. The scaffold collapse was contained within the site boundary. Three workers, who were on the scaffold, sustained multiple injuries. Sadly, one worker died three days later in hospital.

The HSE issued a safety alert to the construction industry following this incident. The warning aimed to alert those working on similar projects to the importance of their arrangements to provide and maintain stable scaffolds. The HSE recommended that those arrangements are reviewed regularly.

The HSE principal inspector said:



It is totally unacceptable for companies to disregard the safety of their workers. If the scaffolding had been designed, erected and managed properly, this incident would never have happened.



Near-fatal scaffold collapse

Three construction firms were fined a total of £40,000 following a near-fatal scaffold collapse on an Essex building site.

Workers at the former Riverside Centre in St John's Avenue, Braintree, were forced to leap to safety through open windows when the 40-metre long and five-storey high scaffolding fell.

Chelmsford Magistrates' Court heard how the Construction (Design and Management) Regulations 2015 (CDM) co-ordinator had not provided suitable advice to the builders, who then failed to implement the requirements of the construction phase plan and monitor the scaffolding works as they progressed.

The scaffolding firm admitted failing to plan the work or design the scaffold. They also failed to send trained and competent workers to the site and, in the weeks prior to the collapse, allowed a scaffolding labourer to act as the supervisor and oversee trainee scaffolders.

Following the collapse, the HSE served a prohibition notice which required that a scaffold design be provided, but the firm continued to send untrained scaffolders to the site and the principal contractor allowed them to continue to work.

The court was told that the principal contractor had received an inspection from the HSE at the same site several months before the collapse, when five enforcement notices were issued for other management failings. A Notification of Contravention was also issued on the CDM co-ordinator for failing to provide suitable and sufficient advice to the client.

The principal contractor pleaded guilty to breaching Regulation 22(1)(a) of CDM. The contractor was fined £20,000 and ordered to pay costs of £2,893.

The scaffolding firm pleaded guilty to breaching Regulations 4(1)(a), 5 and 8(b) of the Work at Height Regulations. They were fined a total of £15,000 and ordered to pay costs of £1,981.

The CDM co-ordinator pleaded guilty to breaching Regulation 20(1)(a) & (b) of CDM. They were fined a total of £5,000 and ordered to pay costs of £1,981.

After the hearing, a HSE inspector said:



This collapse was entirely preventable and it is only by chance that multiple fatalities did not occur. It beggars belief that, following the collapse, no lessons were learnt and untrained people were still allowed to adapt the scaffold.

This case highlights the importance of ensuring those who undertake construction work have the relevant skills, knowledge, training and experience to do so.

Note: whilst every effort has been made to provide reliable and accurate information, we would welcome any corrections to information provided by the writer which may not be entirely accurate; therefore and for this reason, the NASC, or indeed the writer, cannot accept responsibility for any misinformation posted.

CONTENTS

Fall arrest and suspension equipment



5.1	Introduction	76
5.2	Important points	76
5.3	Safety nets	77
5.4	Other soft-landing systems	81
5.5	Personal fall protection equipment	81
5.6	Rope access (abseiling)	87
5.7	Temporary suspended access equipment	87
Appendix A – Inspection of webbing harnesses and lanyards		88
Appendix B – Fall Arrest Safety Equipment Training (FASET) top 10 dos and don'ts for safety net use		89



GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

FALL ARREST AND SUSPENSION EQUIPMENT

Overview

Where a risk assessment has identified that the risk of falls from height cannot be eliminated by the installation of working platforms with barriers, guard-rails or other similar collective measures, the use of personal fall protection or fall-arrest equipment may then be the only option.

Safety nets or the use of other soft-landing systems are preferred to the use of safety harnesses and lines, as they provide a larger area and collective protection for persons working above them. They do not rely on individual workers wearing a safety harness and lanyard connected to a secure anchorage point.

5.1 Introduction

In situations where people and traffic pass below others working at height, in addition to exclusion zones (or danger areas), a safety net used in conjunction with a fine-mesh debris net will help to protect those below from falling objects and materials. It will also provide collective fall-arrest for the people working at height.

Where safety nets cannot be rigged for any reason, and it is not practical to use another form of soft-landing system, it may be necessary to resort to the use of a safety harness and lanyard.

In the first instance work or **fall restraint** should be considered (a technique used that positions the worker so a fall cannot occur). If the worker is required to move around an area where they have the potential to fall, then **fall arrest** should be used (a technique used to reduce the distance and mitigate the consequences of a fall).

- It is imperative that operatives have been trained in the use and care of the equipment provided and that they wear it correctly.
- When operatives are utilising fall-arrest equipment, consideration must be given to the clear space beneath the work area required to allow the energy absorber within the lanyard to operate effectively. A 2 m lanyard anchored at foot level will require approximately 6.75 m of clear space beneath the anchorage point to prevent injury. This distance is made up of the 2 m lanyard length, 1.75 m energy absorber maximum payout length, approximately 2 m body height and a 1 m safety margin.
- A secure, designated anchor point must be available.
- The person working at height must physically attach their personal fall protection equipment to a secure, designated anchor point.

Whichever system is used, whether it is safety nets, another soft-landing system or harness and lanyard, the system must be designed to provide a safe system of work, be installed by competent persons and maintained, inspected and adequately supervised to ensure it is used correctly.

For further information also refer to:



- Chapter D01 Work at Height Regulations 2005
- Chapter D03 Common access equipment
- Chapter D04 Scaffolding
- Chapter F05 Working over or near to water.



In the first instance, where practicable, falls should be prevented by the use of collective control measures. These include the use of edge protection, parapets, scaffolding, temporary towers and mobile elevating work platforms (MEWPs) – all of which are preferable to the use of personal fall protection equipment.

5.2 Important points

- In situations where the prevention of falls from height cannot be guaranteed, it is essential that measures are put in place to ensure that any fall that does occur is arrested, without injury to the person who has suffered the fall.
- In many cases safety nets are the preferred method of arresting falls because:
 - they provide what is termed *collective fall protection* (control measures that, once installed, require no input from those they protect and which can protect more than one user at a time)
 - if rigged immediately below the work area, they reduce the fall distance and can make recovery of anyone who has fallen into the net easier. Recovery should be executed in accordance with the work at height rescue plan
 - they are a soft-landing system that should cause no injury to the person who has fallen.
- The condition of safety nets will deteriorate over time so the requirement to carry out inspections to identify wear and tear will take on additional importance.

- Other forms of soft-landing systems (such as airbags or beanbags) also offer collective protection and may be appropriate in some circumstances.
- If fall protection is to be achieved using a safety harness and lanyard, the wearer must be trained in inspecting, fitting and adjusting the harness and selecting the appropriate lanyard (such as fall restraint or fall arrest) and secure anchor point.
- A harness and lanyard offer individual protection only and therefore require a high degree of personal discipline to ensure they are used correctly.
- Some items used in a fall protection application (for example, retrieval devices) may need a regime of inspections and servicing in accordance with manufacturer's instructions and possibly thorough examination under the Lifting Operations and Lifting Equipment Regulations 1998 (LOLER).
- It is imperative that any personal fall protection equipment is used only in the application for which it is intended. Inappropriate use can make some activities more hazardous and should be avoided at all costs.
- Personal suspension equipment (such as rope-access equipment) must only be used by people who have been trained and are competent. BS 7985 gives practical advice and recommendations for good practice to those who use specialist rope access methods for work at height and those who commission rope access work. Rope access training is available through the Industrial Rope Access Trade Association (IRATA).

[www](http://www.hse.gov.uk) For further information visit the HSE website.

5.3 Safety nets

The use of nets has become widespread throughout the industry, particularly on new build steel frame structures. They have revolutionised safety in roof work. Figures quoted by the trade association FASET (Fall Arrest Safety Equipment Training) suggest that upwards of 50 falls per year are arrested by nets, borne out of the fact that the fatal accident rate in industrial roofing has been massively reduced in recent years as the use of nets increases.

It is important to note that the use of nets does not eradicate risk entirely. It is vital that site managers appreciate that they have an important role in the safe provision of nets on their sites; this is best done by making sure that the net provider is following guidance in British Standards and also guidance produced by the trade body for safety net installers (FASET).

Nets are an area where complying with good practice means testing, inspection, record keeping and storage facilities are required to ensure that the net itself is fit for use. It is important that the end users appreciate that a safe net is not simply about whether the net erectors have a training record card. Systems should also be in place to ensure that the net itself has been checked and inspected.

There are a number of British Standards relating to safety nets. Nets should be manufactured to the requirements of BS EN 1263-1 and erected in accordance with BS EN 1263-2. This latter standard gives information on the installation and use of safety nets.

A further standard, BS 8411, contains construction-specific information and, more importantly, a list of duties for the parties usually involved in construction projects. This includes principal designers, designers, engineers, principal contractors, scaffold contractors and anyone having on-site responsibility for the integrity of safety net installations.

BS 8411 specifies that anyone who is planning the installation of a safety net system should take the following points into account.

- Experience and competence of the net erectors.
- Sequence and type of work being carried out during installation and removal.
- Sequence of construction work to be carried out whilst the nets are in position.
- Provision of effective anchorages.
- Means of access for erecting and removal.
- Access for inspection, debris removal and temporary repair.
- Clearance distances below the net.
- Protection of anyone below.
- Recovery of anyone who has fallen into the net.



Safety nets installed prior to roof works commencing

FALL ARREST AND SUSPENSION EQUIPMENT

Modern safety nets are efficient at saving lives and preventing injury. They are an energy-absorbing system designed to minimise the consequences to the person who has fallen.

Safety nets should be erected as close as possible to the working level to minimise the height of any fall that may occur. The area in which they are installed should be free from obstruction (such as plant or machinery) that may injure a casualty falling into the net.

There are two types of net manufacture.

1. **Knotless**, which provide energy absorption by permanent plastic deformation (stretching) of the net material.
2. **Knotted**, which is generally a heavier and older type of net, providing energy absorption by tightening at the knots and permanently deforming.

Safety nets are manufactured in square or diamond mesh, with two mesh sizes: 60 mm and 100 mm. The 100 mm is the normal mesh size used in the UK.

All safety nets should carry an identification label. This includes the date of manufacture, the net type, class and size, and reference to BS EN 1263-1. It should also carry a unique serial number for record purposes and traceability.

Safety nets should have a tag showing their repair history and any net that is older than 12 months should have a tag showing that it has been tested within the last 12 months.

5.3.1 Fall heights

The positioning of a safety net system is critical to minimise the height of falls that may occur. Safety nets are designed for a maximum fall height of 6 m from the work position, which means that the maximum nominal fall height from the centre of gravity of a person is 7 m.

It should be noted that for safety nets less than 35 m² in total area, or with a side length less than 5 m, the maximum height an operative is allowed to work above a net is 2 m, as there is insufficient fabric area to absorb the energy of falls from a greater height.

However, in accordance with the Work at Height Regulations any fall distance into a safety net must be as little as is reasonably practicable, as the greater the fall distance into a net the greater the likelihood of injury.



When nets are installed, the maximum amount of sag in the net should be no more than 10% of the bay width.

When a load or person falls into a correctly erected net, the net material will deform as it absorbs the energy from the fall. It is therefore critical to provide adequate clearance below the net, to allow the deformation to occur without the load or person striking the ground or another object below the netted area.

Guidance on minimum clearance below nets is available from FASET and the manufacturer's data (for example, if a person were to fall 2 m into a net between 5 m and 9 m wide, the total deformation, including the erection sag, may be between 2.6 m and 3.5 m, depending on the width of net).

It is essential to check the manufacturer's specification to ensure that there is adequate clearance below the planned net position.



Where safety nets are rigged so that the fall distance is less than 6 m, the safety net must extend a minimum of 3 m beyond the leading edge where operatives are working. If the working area is inclined by more than 20° then there must always be 3 m minimum beyond the leading edge.

5.3.2 Competence

The way in which safety net systems are installed is critical. Not only must those installing the net system be trained and competent, so must the people who carry out the routine inspection of safety nets.

After rigging, the nets must be inspected before use by a competent person and a handover certificate should be issued.

The minimum competence someone requires to issue a handover certificate would be a safety net rigger who holds a blue, skilled worker Construction Skills Certification Scheme (CSCS) card, signifying that they have the appropriate experience and training.



The CSCS safety net rigger card is the only evidence that an operative has undergone extensive training and assessment in the proficiency of fall arrest safety equipment training and has achieved the required industry standard.

Always receive a handover certificate from the riggers for each section of netting as it is completed.

There are industry-agreed standards and qualifications for training in the rigging and inspection of safety nets. The training of inspectors is aimed at site management staff as well as professionals within the industry.



Information on netting training providers is available from FASET, the netting industry trade body.

5.3.3 FASET guidance

FASET's website contains the latest technical and safety information, guidance and bulletins for the selection, installation and use of safety nets. These are freely available on their website and should be consulted before selecting and installing nets (for example, understanding the maximum numbers of workers allowed above a safety net, and how to attach safety nets to temporary edge protection).

Items included in the guidance are shown below.

1.	Why specify a FASET member?
2.	Safety net test records.
3.	The testing of safety nets for ultraviolet (UV) light degradation.
4.	Safety net labels and record keeping.
5.	General arrangement for repairing both knotted and knotless safety nets.
6.	Repairs to a knotless safety net.
7.	Repairs to a knotted safety net.
8.	Tagging of safety net repairs.
9.	On-site safety net temporary repair.
10.	Maximum permissible gaps.

11.	Gathering and under-rolling.
12.	Joining safety nets together.
13.	Testing requirements for safety net attachment devices.
14.	Drilled fixings.
15.	Incorrect safety net support clamps – single Gravlock.
16.	Attaching safety nets to temporary edge protection.
17.	Working above a safety net.
18.	Loading of safety nets.
19.	Temporary works.
20.	Ground conditions.
21.	Safety net handover and in-use inspections by a competent person.



FASET's *The selection of access methods to install and dismantle safety netting* can be freely downloaded from its website and site managers should familiarise themselves with it in order to manage and monitor the activities of the installation team.

There are six methods of access currently recommended for rigging and de-rigging safety nets, which should be considered in the following order.

FASET recommended hierarchy for work at height

1.	Rig/de-rig safety nets remotely (using remote attachment devices).
2.	Rig/de-rig using powered access (MEWPs).
3.	Rig/de-rig using ladders (recommended maximum height 4.5 m).
4.	FASET specialist rigger – employing industrial climbing access techniques.
5.	Rope access techniques – IRATA.
6.	Mobile access towers.

Site conditions often dictate that a combination of any or all of the above methods may be required.



Rigging and de-rigging

FASET does not recommend the use of scaffold towers or hop-ups for the rigging and de-rigging of safety nets under normal rigging conditions. There may be rare and isolated occasions for specific work where towers may be appropriate, having taken due regard of the FASET *Good practice guide: the selection of access methods to install and dismantle safety netting* and the hierarchy set out within it. Where such occasions arise, the rigging contractor must prepare a suitable, specific risk assessment taking account of the rigging conditions and the additional control measures required. Towers must always be erected in accordance with the manufacturer's/supplier's instructions by trained personnel.

5.3.4 Periodic testing

Safety nets are typically manufactured from polypropylene or nylon, which are materials that are adversely affected by ultraviolet (UV) light. As such, safety nets must be tested to ascertain whether the safety net meets or exceeds the manufacturer's withdrawal from service level (minimum strength).

All safety nets that are in service must have been tested within the previous 12 months to ensure they meet or exceed this level. These tests typically include removal of a section of mesh and testing it to destruction, to ascertain its structural properties and to ensure the remainder of the net is in good condition.

FALL ARREST AND SUSPENSION EQUIPMENT

Test results must be maintained electronically and should include detail relating to the date, serial number, test result and manufacturer. If this information is not available, the net cannot be considered safe for use.

Nets that have been tested and are safe for use must be tagged, to include the following details.

- Testing company's name.
- Date of the test.
- Confirmation that the test was passed.
- Date the next test is due.
- Any additional information.



For further information refer to FASET's *The testing of safety nets for UV degradation* (Bulletin no. 3).

5.3.5 Inspection

Safety nets are classed as *work equipment* by the Work at Height Regulations and therefore, in addition to pre-use or handover inspections, they should be inspected on a weekly basis by a competent person to ensure that they are still in a safe condition, fixed correctly and will provide the fall arrest capability if required.

Inspections should be carried out more frequently if circumstances indicate that the integrity of the net system is in doubt.

A net should also be inspected after a person or substantial load has fallen into it, to determine whether it should remain in service or be replaced. In some cases, it may be necessary to seek specialist advice.

It is recommended that the findings of all inspections are recorded.



A safety net inspection sheet template is available from FASET upon request.

5.3.6 Care of nets

Care should be taken to reduce to a minimum unnecessary wear and mechanical damage likely to weaken the net.

Materials must not be stacked on it, and deliberate jumping into it or dropping of objects onto it must be prohibited, as permanent deformation may occur.

The following sources of damage or wear should be avoided as far as possible.

- Dragging the net over rough surfaces.
- Contact between the net and sharp edges.
- An accumulation of debris in the net.
- Any sparks from hot work, welding, grinding, burning operations, hot gases from blowlamps, or hot ash from chimneys or furnaces.
- Chemical attack.
- Any form of radiation.

Special care should be exercised and precautions taken to prevent the net and any supporting framework from being struck by loads on moving vehicles or by the vehicles themselves.

The net manufacturer should be consulted when there is any doubt about the suitability of nets for use in hazardous conditions, after any known contamination or when deformation has occurred.

If overhead power lines are present in the vicinity of an area where safety nets are required, they should not be installed (nor should any access to height be permitted) until such time that the electricity supply company has been contacted and it has been established that it is safe to do so.

5.3.7 Maintenance

Nets must always be thoroughly inspected after use and before being returned to storage to identify any damage and any required repairs.

If contaminated by acids or alkalis, nets should be thoroughly washed, preferably by hosing, and allowed to dry naturally away from heat.

If areas of damage are found, or chemical damage is suspected, contact the manufacturer to obtain a list of competent people able to repair or clean the nets.

5.3.8 Storage

- Nets should be stored away from heat, chemicals and out of direct light.
- Nets should be stored in dry conditions.
- Nets should be stored away from rodents to minimise damage.
- Wet nets should be dried naturally.
- Storage areas should be well ventilated.
- Nets should be turned periodically to allow air circulation.
- If stacked, nets should be packed up clear of the ground.
- Nets should be segregated into three distinguishable categories (for example, nets that are safe for use, nets that are awaiting repair and nets that are beyond economical repair).

5.3.9 Rescue from a net system

Where the net is erected as close as possible below the work area, many of the situations where persons enter a net will be minor, with a self-rescue being possible. On other occasions, a person may fall a considerable distance into a net. They may fall onto materials lying in the net, or strike their head or body on, for example, structural steelwork during the fall.

When such accidents occur, extreme care must be taken during the rescue of the person lying injured in the net. Due to the stretching nature of the net, it is possible that any rescuer entering it could inadvertently and unavoidably cause further injury to the victim.

It is therefore essential that companies using safety net systems have, as part of their risk assessment process, emergency procedures written for:

- treating first aid needs whilst the injured person is in the net
- emergency recovery from a rigged net system (an additional two people may enter the net in an emergency situation)
- gaining access to height to execute a prompt recovery of the casualty.

It must be recognised that procedures for the above must be robust and, where additional equipment is required (for example, MEWPs), this should be instantly available in the work area and suitable for the circumstances. Therefore, rescue must always be pre-planned and must consider the work activities and the changing site environment.

5.4 Other soft-landing systems

Soft-landing systems can provide an effective alternative to safety nets in some circumstances. They are essentially bags that are packed with polystyrene chippings or another energy-absorbing material, which serve to both cushion a fall and reduce the distance of that fall (by the depth of the bag). The bags are linked together with plastic snap clips to completely fill the area over which protection is required. They can also be used on the first or subsequent floors while trusses are being installed or in the roof space when fixing bracings.

An alternative system that may be considered in appropriate circumstances is the use of air-filled bags. Similar to the above, bags of varying sizes may be clipped together to completely fill the area over which fall-arrest protection is required. Airbags require an air compressor running all the time that fall arrest is required, to maintain the pressure in the airbag system. These devices work on the principle of a controlled rate of constant inflation and leakage so that the airbags will absorb the energy of someone falling onto them without bouncing. If air bags are used at first floor level measures must be put in place to prevent any person falling into the air bags from being catapulted out of open window reveals. This can occur if the bags are over inflated or packed too tightly against walls in the fall zone.

Whilst soft-landing systems do not prevent falls, they can be effective in eliminating injuries arising from low-level falls. However, care should be taken when selecting the particular system for use as maximum fall heights vary from product to product.

5.5 Personal fall protection equipment

- All personal fall protection equipment (PFPE) must be manufactured to the relevant British and European Standards.
- PFPE must only be selected if safer options cannot be justified.
- The use of PFPE must be justified by a risk assessment. (Methods of rescue must be considered.)
- All training must be carried out in accordance with BS 8454 (the *Code of Practice for the delivery of training and education for work at height and rescue*).
- All operatives using PFPE must be trained in the use and inspection of PFPE. Training must also include specialist equipment required for rescue.

FALL ARREST AND SUSPENSION EQUIPMENT

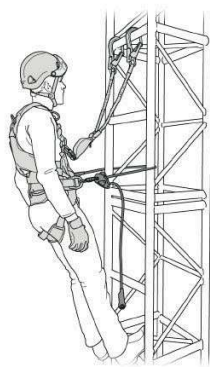
- All PFPE must be subject to a pre-use inspection by the operative, which must be included in the user training.
- Periodic examination of equipment must be carried out by a competent person who has received specific training.
- Contractors must compile and retain records of examination as recommended by HSE guidance INDG367.
- PFPE must be identified by its unique serial number or other means.
- Any PFPE found to be defective or damaged must be removed from service immediately and quarantined to prevent it from being used.

If fall prevention measures (for example, working platforms, barriers or guard-rails) or collective fall-arrest measures (such as safety nets or other soft-landing systems) are not practical, an alternative safe system of work must be employed. This safe system may require the use of safety harnesses and lanyards but, as with any item of personal protective equipment (PPE), it should be considered only as a last resort. Safety nets and other soft-landing systems are collective measures (in other words, they automatically provide protection for more than one person), whereas safety harness systems protect the user only. PFPE can be broadly grouped into the following four categories.

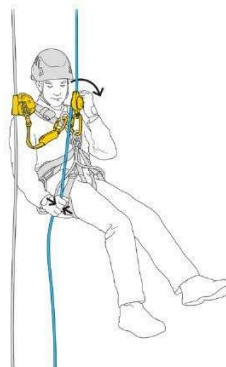
- Fall-restraint systems.
- Work positioning systems.
- Rope access systems.
- Fall-arrest systems.



Fall-restraint system (Photo © NASC)



Work positioning system
(Image supplied by PETZL)



Rope access system
(Image supplied by PETZL)



Fall-arrest system (Photo © NASC)

5.5.1 Selection of equipment

It is important to recognise that few safety harnesses serve multiple purposes and they are often provided for use in a particular application. Examples include harnesses that arrest falls, harnesses that restrict movement of the individual to prevent access to the leading edge and harnesses that are used to facilitate the rescue of an individual from a dangerous situation (such as a confined space or vessel).

Anchor points and fall distances are critical considerations when planning to use a safety harness and lanyard. Consequently, individuals planning work of this nature should have a high degree of training and competence and the work activity will need to be closely supervised.

The type and nature of the safety harness to be used must be provided only after having given consideration to the individual who will be using it, the work activity to be carried out and other factors that may impact on the effectiveness of the equipment (for example, the ability of a lanyard to resist fraying or cutting from a sharp or abrasive edge in a fall, known as a running edge).

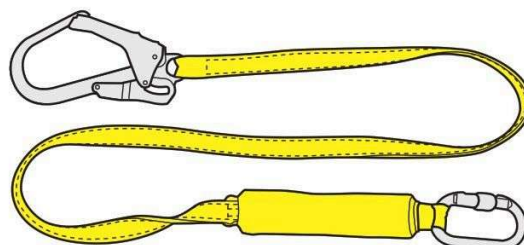
In addition to different types of harness, there are also several types of lanyard, with each being intended for a particular purpose. Some of these are discussed below.

5.5.1.1 Fall-arrest lanyards

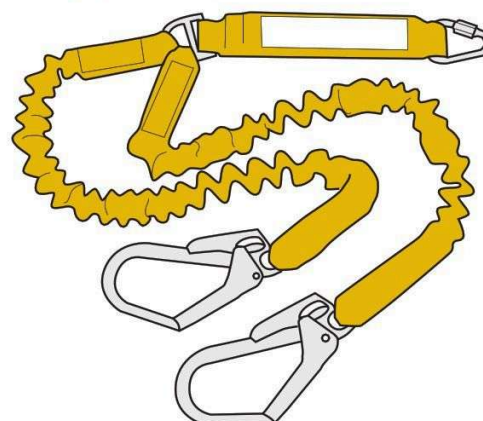
Fall-arrest lanyards incorporate an energy-absorbing feature to reduce the shock loading on the body of the wearer.

5.5.1.2 Twin tailed lanyards

Twin tailed lanyards are a type of fall-arrest lanyard. They allow greater mobility at height by enabling the repositioning of one tail at a time so that the user is constantly attached to an anchor point.



Fall-arrest lanyard (Photo © NASC)



Twin tailed lanyard (Photo © NASC)

FALL ARREST AND SUSPENSION EQUIPMENT

Users of this type of lanyard should note that if used incorrectly they can pose additional risks to safety. If only one tail is clipped to a secure anchorage and the second tail is not located correctly, then during a fall the second tail could loop over a fixed object and arrest the fall before the energy-absorber has deployed, which may cause severe personal injury.

The second tail must never be clipped back on to the user's harness, unless it is fitted with purpose fitted (sacrificial) parking points that will break away from the harness if the second tail comes under tension. Alternatively, the second tail can be either left to hang free or be simultaneously clipped to the same secure anchorage.

5.5.1.3 Restraint lanyards

Restraint lanyards are used as a means of limiting the range of movement of the wearer in order to stop them falling from an exposed edge. For example, when allowing maintenance to be carried out on a gutter on the edge of a roof, the worker can reach the gutter but the lanyard is not long enough to allow the worker to physically lose contact with the structure, thus preventing a fall from the roof.

These lanyards are not designed to arrest falls and have no energy-absorbing feature. Sometimes two of these may be worn at the same time, with each lanyard fixed to a clip on either side of the harness (EN 358) to make the harness what is termed a work positioning harness.

Irrespective of the type of lanyard used, they are only effective if the free end is securely anchored to a suitable anchorage point.



Restraint lanyard (Photo © NASC)

5.5.2 Training

Any training should cover the selection, fitting, adjustment, maintenance and use of personal fall protection equipment and explain the choice and use of suitable anchorage points. Employees should not be permitted to use the equipment before adequate instruction has been given and they have been judged to be competent in its use.

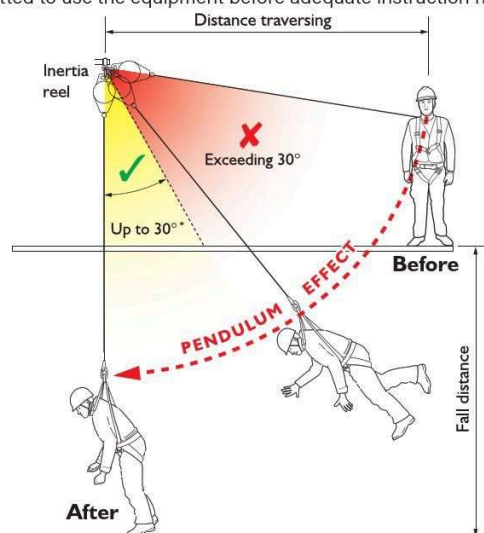
5.5.3 Selecting the anchor point

When working in fall arrest, in order to limit the drop, the designated anchorage points should always be as high as possible above the person and as near to vertical as possible in order to avoid the pendulum effect (the casualty swinging from the point of work to the point of attachment).

Anchor points must be capable of withstanding the anticipated shock loading (refer to *Personal fall protection equipment. Anchor systems. System design, installation and inspection. Code of Practice (BS 7833) and other industry guidance*).

Consideration should also be given to how persons would be rescued following an arrested fall, particularly when work is from high structures.

Some harness manufacturers also produce rescue systems that enable a single rescuer to raise a suspended person back to the working platform or safely lower them to ground level, or self rescue for a person who is suspended but conscious and uninjured to safely lower themselves to ground level.



Pendulum effect

5.5.4 Arrestor devices

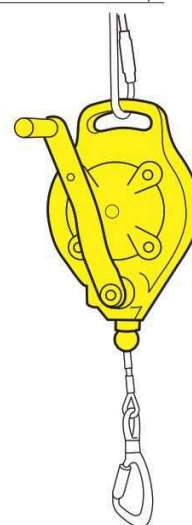
These devices operate with a similar mechanism to the seat belts used in cars and snap to the locked position as a result of sudden movement.

Most inertia reel-type arresters are only designed to safely arrest someone who is working in and has fallen from a position that is almost directly vertical to the anchor point. As with lanyards, these devices will not counter the pendulum effect or swing fall.

Most equipment of this nature is not tested for over the edge type falls in which the retractable lanyard is pulled tight across the edge of a surface (for example, a roof sheet or floor slab) by the weight of the fallen person. In these circumstances, the abrasive action of webbing being pulled taut over an adjacent surface can have catastrophic effects in terms of the device's ability to successfully protect against falls.

If the proposed work method involves the equipment trailing over a running edge, it is unlikely to be a suitable method of work and advice must be taken from the manufacturer to ensure that the equipment is suitable for use in this application.

Other types of fall arrester are available, such as a traveller on a pre-tensioned vertical cable. There are also proprietary systems where the lanyard is attached to a traveller that moves along a pre-positioned and tensioned horizontal or vertical steel cable.



Typical arrester device (Photo © NASC)

FALL ARREST AND SUSPENSION EQUIPMENT

These systems can improve movement around corners and past obstacles without the need to unclip. Advice should be sought on the selection of the best type of equipment to be used for the job to be done. Refer to the manufacturer for information about inspection and maintenance of fall-arrest devices. Where the client provides this equipment, as is common in many cases, care still needs to be taken that it has been inspected and checked as required and that the users are trained and understand how it works.

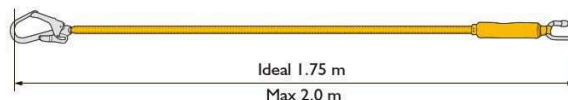
5.5.5 Shock absorbers

If a person wearing a harness and shock-absorber lanyard falls, there is a considerable shock loading to the body impact force; the greater the fall distance, the greater the force that is applied.

The maximum distance a person can fall will depend on the height of their anchor point and length of lanyard. A higher anchor point essentially means the fall distance is reduced.

In a worst-case scenario, with a 2 m lanyard (the maximum permitted length under EN 354) anchored at foot level, the area of clear space beneath the person required is 6.75 m, made up from the following.

- 2 m lanyard length.
- 1.75 m energy absorber extension.
- 2 m body height.
- 1 m safety.



A shock-absorbing lanyard (Photo © NASC)

Shock absorbers in the form of tearaway stitching, stretch springs or a deforming metal strip are built into fall-arrest lanyards as a means of reducing the shock loading.



A lanyard that has been used to arrest a fall should never be reused. It should be withdrawn from service immediately and destroyed or returned to the manufacturer.

5.5.6 Markings on fall protection equipment

Fall protection equipment must be clearly and indelibly marked or permanently labelled with the following.

- British or European Standard to which it conforms. Check equipment is compliant to the current standards.
- Name, trademark or other means of identification of the manufacturer.
- Year in which the fall protection equipment was manufactured. These have a maximum working life of 5-10 years, depending on the manufacturer.
- Type of fall protection equipment.
- Manufacturer's serial number.
- Company serial number, or other recognition system, for recording maintenance and inspections.



The Work at Height Safety Association (WAHSA) has produced leaflets and information on the use and positioning of fall-arrest equipment for those working at height.



Main performance standards

- *Code of Practice for selection, use and maintenance of personal fall protection systems and equipment for use in the workplace* (BS 8437).
- *Code of Practice for delivery of training and education for work at height and rescue* (BS 8454).
- *Personal fall protection equipment. Anchor systems. System design, installation and inspection. Code of Practice* (BS 7883).
- *Code of Practice for the use of rope access methods for industrial purposes* (BS 7985).
- *Inspecting fall-arrest equipment made from webbing or rope* (HSE guidance INDG367).
- WAHSA technical guidance notes.
- IRATA International Code of Practice.
- *Preventing falls in scaffolding operations* (SG4).



For further information visit the HSE website.

5.5.7 Storage

When not in use, equipment should be stored in a cool, dry place and not be subjected to direct sunlight. The use of purpose-designed cabinets, which allow ventilation, is recommended. If fall protection equipment becomes wet it should be dried thoroughly by natural means.

The equipment should not be subjected to unnecessary strain or pressure and must be kept free from contact with sharp implements, corrosives and other possible causes of damage. Manufacturer's guidance is the best resource for checking usage, storage and maintenance requirements.

5.5.8 Inspection



Reduction of a lanyard's strength

Research carried out for the HSE involving synthetic fibre webbing lanyards confirmed potential causes of degradation. A 1 mm cut in the edge of a lanyard was shown to result in up to a 40% loss of strength, depending on the make of lanyard being used for the test. Most users do not appreciate the possible loss of strength as they assume that there is a proportional strength-to-width relationship; so in their eyes a 1 mm nick on a 25 mm flat sling is not really significant.

A further study investigated the resistance of the webbing material (used to make harnesses and lanyards) to abrasion by various types of surface, including concrete and angle iron. The findings of this study revealed that severe abrasion of the material, and a significant reduction in its strength, occurred after a small number of cycles of abrading the material across concrete. Abrasion across angle iron also degraded the material but at a significantly slower rate.

The wearer must make a visual inspection of safety equipment **before use**. The equipment should be examined by a competent person at least once every six months and the results recorded. Equipment examination should be carried out at the following minimum frequencies.

- Every six months for normal use.
- Interim examination, as determined through risk assessment.
- Every three months for extreme conditions, environments or occupations.
- Installed systems (horizontal and vertical lines), according to the manufacturer's recommendations.

Many contractors have a weekly examination regime because of the rigorous conditions of use.

Harnesses and lanyards should be examined by a competent person after a fall or other circumstances in which the equipment has been deployed, before it is reissued for use. Any equipment found to be damaged should be removed from service immediately.

Safety belts, harnesses and lanyards should be taken out of use if found to be damaged or defective. Under most circumstances, knots in lanyards would be considered a significant problem. Typically, a knot is presumed to reduce the strength of a lanyard by 50%. Particular attention during inspection should be directed to the points below.

Webbing and leather. Examine for cuts, cracks, tears or abrasions, stretching and distortion, damage due to deterioration, contact with heat, acids or other corrosives and rot.

Snap hooks. Examine for damaged or distorted hooks, faulty springs, strained jaws, hairline cracks and corrosion.

Buckles. Carefully examine the shoulders of buckles; inspect for open or distorted rollers, undue wear, hairline cracks and corrosion.

Stitching. Examine for broken, cut or worn threads, open seams and failed or loose stitching.

Lanyards and ropes. Examine for damage or signs of wear, cuts, abrasions, flat spots, knots and permanent kinks. For kernmantle ropes (the variety often used in climbing and abseiling activities), also check for visible signs of the inner core.

Unauthorised modifications. Examine equipment for homemade attachments, repairs or adaptations. Impress on the wearers that their lives could depend on the continued efficiency and durability of their safety equipment and that, by frequent personal inspections, the possibility of equipment failure will be reduced to a minimum.

For webbing lanyards, specific attention should be given to the following.

- Cuts of 1 mm or more to the edge of the lanyard (for example, as a result of being choke-hitched around steelwork).
- Damaged or deformed fittings.
- Surface abrasions to surface or edges and damaged stitching.
- Partially deployed energy-absorbers.
- Knots in the lanyard.
- Ultraviolet (UV) degradation (identified by possible loss of colour and a powdery surface).
- Results of chemical attack.



For full details refer to the HSE publication *Inspecting fall arrest equipment made from webbing or rope* (INDG367).

FALL ARREST AND SUSPENSION EQUIPMENT

5.5.9 Records

An inspection and maintenance record should be kept for each harness and lanyard. This should be used to document the life cycle of the equipment. It is imperative that each piece of equipment should be marked with an individual serial number for identification purposes. If the unique serial number is no longer visible, the item should be withdrawn from service and replaced.

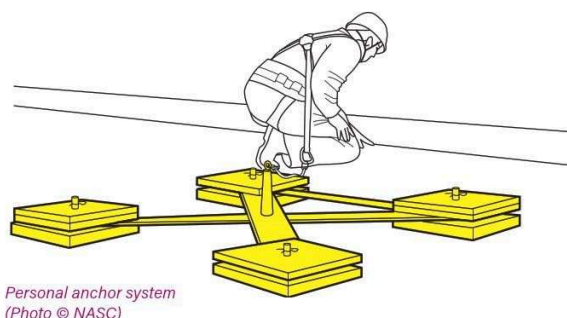
5.5.10 Dead weight anchor devices (mobile person anchors)

The use of dead weight anchors in accordance with European and British Standards BS EN 795 and BS 7883 (Class E) has become an acceptable means of providing a safe fall protection system on flat roof surfaces, particularly where it is not possible to penetrate the roof surface. They can sometimes be used in both fall-restraint and fall-arrest applications, but are more commonly associated with fall restraint.

BS EN 795 requires these devices to be tested for capability to arrest falls due to the foreseeability of misuse, even for use in a restraint application (note that BS EN 795 applies only to single-user systems. Where multiple-user systems are required, PD CEN/TS 16415 will apply).

As with all other systems, deadweight anchors do have their limitations, as detailed below.

- Location distances of the anchor device from the edge of the surface they offer protection against.
- Their safe use is impacted by the presence of water as it will reduce the level of friction available between surfaces.
- They can be cumbersome to install.
- They are less suitable for heavier people (due to loading limitations).
- They are often unsuitable for use on surfaces with a slope in excess of five degrees.



Personal anchor system
(Photo © NASC)

As with other items of fall protection equipment, deadweight anchor devices should be used only after thorough consideration of the task and reference to manufacturer's guidance has been made.

5.5.11 Suspension risk (syncope or fainting)

Many people who work in the fall protection and construction industries often talk about *suspension trauma* but HSE research (RR708) suggests the term *suspension trauma* is inappropriate because orthostasis, motionless vertical suspension is responsible, as opposed to physical injury (trauma), as was widely believed. For this reason, the terms *suspension syncope* or *fainting* are preferred.

Previous guidance recommended that rescued casualties should be placed into a sitting position, as opposed to the more common recovery position suggested by most medical guidance.

However, the HSE's research document states: 'there is no evidence to support assertions that, post-rescue, this position is physiologically superior or even equivalent to the horizontal position' and consequently the traditional lying-down recovery position is recommended.

In all situations, when a worker has fallen and their fall has been arrested by fall protection equipment, the individual should be retrieved as quickly as possible to prevent the onset of fainting. In many cases, the casualty may be able to assist either fully (self-rescue) or partially.

Casualties who are conscious and uninjured and are suspended by fall protection equipment are able to slow the onset of suspension syncope by moving their legs, by positioning their body in a horizontal or slight leg-high position or, in the presence of the appropriate equipment, using a proprietary suspension loop that will enable the casualty to stand up whilst suspended.

When rescued, the casualty should be referred to a medical practitioner for professional medical assessment, even if it seems there have been no harmful effects caused by the fall.



Research suggests that if rescue is effected within 10 minutes, most people will not experience symptoms of pre-syncope (dizziness) or syncope (fainting).

5.5.12 Harness fall recovery with specialist equipment

Many of the companies who manufacture and supply fall protection equipment also produce kits that are specifically designed to facilitate the rescue of those suspended at height.

Depending upon the design of the equipment, sometimes a single person is able to attach the rescue system to the D-ring of the fallen person's harness and either lower them to ground level or raise them back up on to the working platform in a controlled manner.

This type of rescue equipment is classified as lifting equipment under LOLER and therefore carries all of the training, examination and maintenance obligations that this brings.

5.6 Rope access (abseiling)

Rope access is often seen as a cost effective solution for certain work at height activities. However, due to the risk involved and high levels of skills and control required, it should only be selected after all other forms of collective protection have been considered.

It is generally suitable for inspection, cleaning and maintenance work, but is only used for construction activities in exceptional circumstances. An example might be the rigging of safety nets where it is not reasonably practicable to do it by using other means of access (such as a MEWP or another form of working platform).

General requirements for rope access are listed below.

- All personnel involved in rope access operations must be trained and assessed in accordance with the IRATA training scheme.
- Each working party (crew) must have at least one Level 3 technician who is responsible for the direct planning and supervision of the work.
- The team leader shall prepare an access permit before beginning rope access work. The access permit should include all salient aspects of the work to be undertaken, including detail of other workers involved, equipment to be used, associated hazards, PPE requirements, rescue arrangements and safety arrangements (safety line and back-up line), and so on.
- All work must be carried out in accordance with the following.
 - The *Code of Practice for the use of rope access methods for industrial purposes* (BS 7985).
 - *Personal equipment for protection against falls. Rope access systems. Fundamental principles for a system of work* (ISO 22846 Part 1).
 - The *Code of Practice. Personal equipment for protection against falls. Rope access systems* (ISO 22846 Part 2).



IRATA is the trade association that governs the rope access industry.

A second group of specialists (steeplejacks and lightning conductor engineers) commonly use rope access, boatswain's chairs and bosun's seats. Steeplejacking is work at height and therefore the requirements of the Work at Height Regulations apply, as they would to any other work at height activity.



The trade associations for the steeplejack industry are the Association of Technical Lightning and Access Specialists (ATLAS) and the Steeplejack and Lightning Protection Training Group (SLPTG).

5.7 Temporary suspended access equipment

Temporary suspended access equipment can refer to a wide range of equipment that is used to support or contain workers working at height.

It includes equipment such as a single suspended chair (for example, a boatswain's chair), usually suspended from the roof and suitable for one person carrying out light work (such as for window cleaning or working on the side of a building), and a platform, cradle or work cage, suspended from the roof, that can have a load rating of 1,000 kg (or more) and a span of more than 15 m, capable of carrying a number of persons.

These larger types of equipment are typically used in industrial and commercial construction projects (such as building, restoration, painting, sandblasting, waterproofing, window cleaning, inspections and general building maintenance).

The equipment is a temporary means of carrying out a specific task; the equipment is dismantled and removed from the work area upon completion of the task.

Temporary suspended access equipment is a specialist subject and is covered by the British Standard for **Planning, design, setting up and use of temporary suspended access equipment - Code of Practice** (BS 5974:2017).

The equipment is normally raised or lowered by means of a manually operated or powered hoist or climbing device that can be attached to the temporary suspended platform, work cage or suspended chair.

Permanently installed suspended access equipment is not covered here or within BS 5974:2017.

For trade association information relating to subjects that do not fall under the temporary suspended access equipment standard (for example, rope access (IRATA), Steeplejack and Lightning Protection Training Group (SLPTG) and Association of technical lightning and access specialists (ATLAS) refer to 5.6.



For further information visit the Specialist Access Engineering and Maintenance Association (SAEMA) website.

FALL ARREST AND SUSPENSION EQUIPMENT

Appendix A – Inspection of webbing harnesses and lanyards

The HSE and other key industry stakeholders (including the British Safety Industry Federation, the Height Safety Group and the Working at Height Safety Association) recommend that a three-tier inspection regime (pre-use, detailed inspections and interim inspections) is implemented where webbing harnesses and lanyards are in use. The regime should be drawn up by a competent person and include the following information.

- The equipment to be inspected (including unique identification references).
- The frequency and type of inspection (pre-use checks, detailed inspection and, where appropriate, interim inspection).
- Designated competent persons to carry out the inspections.
- Action to be taken on finding defective equipment.
- Means of recording the inspections.
- Training of users.
- Means of monitoring the inspection regime to verify inspections are carried out accordingly.

All persons who carry out any inspection must have the independence and impartiality to make objective decisions, and sufficient authority to discard defective equipment.



In the unlikely event of a fall, a webbing harness and lanyard that are worn out or damaged may not take the impact and stress of a person arresting, and fail. A harness, lanyard and secure anchor point are the last resort to a person falling – their sole purpose is to save lives.

The importance of pre-use checks and regular inspections cannot be overemphasised.

Inspection regime

Harnesses and lanyards should be subjected to the following checks and inspections.

Pre-use checks

These are visual and tactile inspections, carried out by slowly passing the lanyard or harness through the hands to detect small cuts in the edges, softening or hardening of the fibres and traces of contaminants. The inspection should only take a few minutes and should be carried out by the user prior to each period of use. These inspections do not need to be recorded.

Detailed inspections

These are more formal and in-depth inspections carried out at least every six months, and every three months for frequently used equipment, particularly where equipment is used in extreme conditions or environments (such as demolition, scaffolding and steel erection). They are normally carried out by a designated individual who is also responsible for recording the results and implementing the required action.

Interim inspections

These are in-depth, formal inspections, carried out when it is suspected that equipment has deteriorated and an inspection is needed before the next scheduled detailed inspection. Results of these inspections must also be recorded.



Examples of damaged equipment – the user should check for signs of tears, damage and wear

Appendix B – Fall Arrest Safety Equipment Training (FASET) top 10 dos and don'ts for safety net use

FASET top 10 dos and don'ts for safety net use	
Do	
✓	Ensure you use competent safety net rigging companies with relevant experience (such as FASET member companies).
✓	Ensure you use competent safety net riggers (those with a safety net rigger competency card).
✓	Ensure that methods of access for work at height have been properly assessed, and the safest method that is reasonably practicable has been chosen.
✓	Carry out a formal inspection if nets are in place for more than seven days, or subjected to extreme weather conditions. A visual user inspection should be carried out prior to working above the safety nets.
✓	Ensure that all materials are secure before lifting or craning above a safety net.
✓	Ensure that a handover certificate is completed prior to allowing works above a safety net.
✓	Ensure that nets are struck prior to any hot works being carried out above.
✓	Report any damage to a safety net.
✓	Ensure all falls into a safety net are reported to site management and call FASET. Ensure the supervision of the loaded net's removal from service.
✓	Ensure that the safety nets used conform to BS EN 1263-1:2014 and are rigged in accordance with BS EN 1263-2:2014.
Don't	
✗	Allow the use of nets unless there is evidence of an appropriate inspection and maintenance regime and UV testing within the last 12 months.
✗	Allow the use of nets unless a competent person has verified there is sufficient clearance distance (rule of thumb: no less than 3 m).
✗	Allow work above nets without evidence that any repairs have been carried out by a competent person authorised by the manufacturer.
✗	Allow nets to be installed which don't have serial numbers or labels.
✗	Allow workers to use the nets as a means of access or working platform.
✗	Attach nets to any structure without first confirming it has the capacity to deal with a minimum 6 kN loading at a 45° angle.
✗	Allow gaps greater than 100 mm unless exceptional circumstances dictate otherwise (in these cases, not greater than 225 mm).
✗	Tamper with or adjust any nets and attachment points.
✗	Allow work above net systems that have been damaged or loaded.
✗	Allow work above the net if it is positioned greater than 2 m from the working level.

CONTENTS

Excavations



6.1	Introduction	92
6.2	Important points	92
6.3	Excavation support	94
6.4	Excavation safety	97
6.5	Inspections and reports	100
6.6	Excavators used as cranes	101
Appendix A – Safety awareness around excavators		102
Appendix B – Case study – Excavation collapses after dangers are ignored		104
Appendix C – Case study – Building contractor jailed after house collapse		104



GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



Scan the QR code to complete a brief survey, and share your feedback on the course.

EXCAVATIONS

Overview

This chapter gives practical guidance on factors to be considered when planning excavations, setting up of the work and safe working within them. This includes elements to consider, such as soil composition, depth, support structures, access and egress, vehicles and plant, and guarding excavations.

Good planning and correct preparation for excavations are essential, as every year people are killed or seriously injured by collapses and falling materials while working in excavations.

6.1 Introduction

Most construction work will involve a form of excavation for foundations, sewers or drainage. These will vary in size and depth but are always a high-risk activity as they can be affected by diverse factors (such as rainwater, hot weather, adjoining structures, ground conditions and vibration).

Many accidents occur when the excavation appears to be in good condition, with its sides clean and self-supporting, and with no obvious hazards.

Some excavations may need to be classed as confined spaces, and will therefore need additional precautions. Excavations and excavation supports also require temporary works procedures to be followed. Some excavation and excavation support work may require specific design considerations and additional precautions.



For further information refer to Chapter C05 Temporary works.



Neither the shallowness of an excavation nor the appearance of the ground should be taken as indicators of safety. Accident reports suggest that far too often such assumptions are made incorrectly.



An unsafe excavation

The Health and Safety Executive (HSE) inspector served a prohibition notice for this excavation, which was 1.9 m deep and unsupported.

The contractor was fined £10,000 and the HSE was awarded £8,000 costs.

In addition, the supervisor was fined £1,000 and the HSE was awarded £800 costs.



The method of excavation must be specific to local conditions and may need to be different to an excavation nearby. Workers will often have to improvise and use their knowledge and skills to overcome unforeseen obstructions and different ground conditions. It is important that supervisors and their workers are trained and competent to ensure excavation work is completed in a safe way.

6.2 Important points

A **risk assessment** must be completed for any works that are to be undertaken, and this must be done by a competent person.

This risk assessment must take into consideration the potential for the sides to collapse, that someone may be injured and need first aid, a site-specific rescue plan, and the possible need for a support structure.

The risk assessment should eliminate or reduce the risk at source (such as using trenchless techniques or having the sides battered or stepped). Risks, from the following hazards, must also be considered.

- Collapse of the sides.
- Underground services.
- Contaminated ground.

- Fall of persons, materials, plant or equipment into the excavation.
- Confined spaces – poisonous or explosive atmospheres or lack of oxygen.
- Flooding.
- Overhead services.
- Moving plant – injury to persons.
- Lifting operations.
- Undermining adjacent structures, scaffolds, services or trees.
- Surcharging the excavation, meaning applying pressure to the sides (for example, vehicle routes or material storage areas too close to the excavation).



For details on safety awareness for working around excavators refer to Appendix A.

Control measures described in the risk assessment should include the following.

- Protection of person(s) installing the support system.
- Identifying safe exposure and, if necessary, support of underground services.
- Safe access to and egress from the excavation.
- Adequate ventilation of the workspace.
- Arrangements to dewater the trench if necessary.
- Inspections of the excavation by a competent person.
- Consideration for the stability of adjacent structures or land.
- Arranging for appropriate signing, guarding and lighting and, where necessary, protection of non-employees (pedestrians and the public).



A cubic metre of soil can weigh over a tonne.



A collapse captured while taking site record photos

The excavation was 1.8 m deep and the collapse took less than two seconds. The method was to dig and pour concrete with no-one entering. Due to heavy rain, this method was abandoned and trench support installed. The section that broke away weighed about one tonne and it can be seen how easily a serious accident could happen.



Information on the structural stability of excavations can be found on the construction pages of the HSE website.

EXCAVATIONS

6.3 Excavation support

There will always be water present during excavations, even if only moisture in the soil, and this is an additional hazard to be considered (as shown in the illustration on the right).

Soils that are akin to fine sand flow easily; however, those like stiff clay stick together more easily.

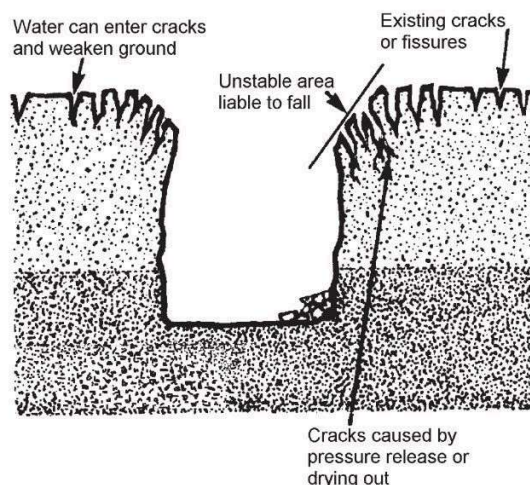
Do not rely on any soil, whatever its structure, to support its own weight. If it cannot be battered or stepped, then alternative support will be essential.

Loose and fractured rock will also need support.

Before commencing any excavation work consider what temporary support is required, and whether any equipment or special precautions are needed.

Clients must provide information to contractors before work begins. This should include relevant information on ground conditions, underground structures or watercourses and the location of existing services.

This information should be used during the planning of, and preparation for, excavation work. It will allow the contractor to choose the most suitable method to support the excavation.



An excavation in firm or stiff clay-type ground



Prevent collapse – shore, step or batter back. NEVER assume the ground will stand unsupported, even for a short time.

6.3.1 Trenchless techniques

Persons planning construction work should be aware of alternative techniques that may be able to partially or totally eliminate the need for excavations.

Examples of the trenchless techniques include boring, directional drilling and pipe jacking – some will need excavation pits at each end to launch and retrieve equipment and to make connections to pipework or jointing of cables.

6.3.2 Battering or stepping excavations

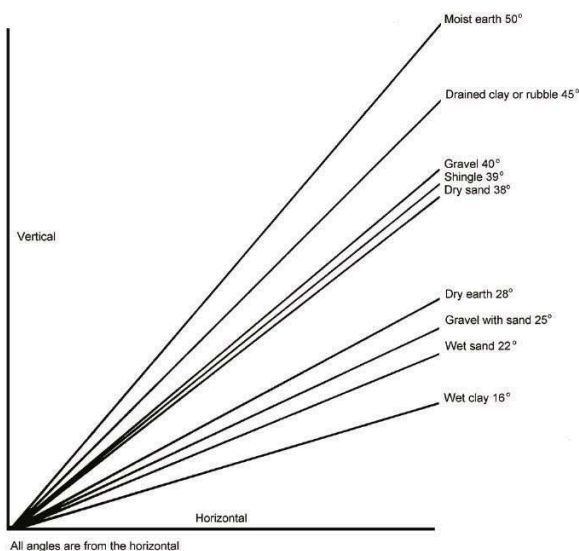
Battering back or stepping the sides of an excavation to a safe angle is an acceptable means of preventing instability and collapse. In many situations this is the safest and simplest way of ensuring stability and should be given due consideration.

The diagram (right) is an example of typical safe slope angles and should be used for guidance purposes only. If there is any doubt regarding soil conditions, the site engineer or the shoring provider should be consulted.

Battering or stepping excavations will depend on the nature of the soil, which could be one material or could be a mix of materials. Consideration will need to be given to the level of the water table and water content, and the increase or decrease of this while the excavation is open. A small percentage increase in water content can reduce the stability of the excavation sides.

The slope angles described in the diagram are known as the **angle of repose**. In granular soils the angle of slope should be less than the natural angle of repose of the material being excavated. This will prevent the granular material from *running*. In wet ground a considerably flatter slope will be required.

Plant, pedestrians, materials and equipment should be kept at a safe distance outside the angle of repose to prevent surcharge (vertical pressure or load that acts over the ground surface) on the sides of the excavation.



All angles are from the horizontal

Typical safe slope angles. The vertical distance should not exceed 1.2 m



Stepped excavation – but showing chapter 8-type barriers, which should not be used for edge protection

This image demonstrates bad practice.

It shows red, plastic-style barriers, which are not suitable for this type of excavation work, but which should be used for street works and road works on the public highway in accordance with *Safety at Street Works: A Code of Practice* (Traffic Signs Manual, chapter 8).

(Image supplied by Barconn Ltd/Paramount Health and Safety)



6.3.3 Trench supports

The type of support will depend on the following.

- The type of excavation.
- The nature of the ground.
- Groundwater conditions.
- Surcharge of the excavation sides.

Depending on the risk assessment the excavation may need to be supported.

A survey should be carried out for all but shallow trenches, to determine the type of soil and other ground conditions; this will then identify what support will be required. This survey must be carried out by a competent person. A specialist engineer will be required for large or complex excavations. In the more complex projects bespoke solutions will be required.

Equipment for support must be available before excavation work is started, and be of good quality and well maintained; they must also be fixed securely.

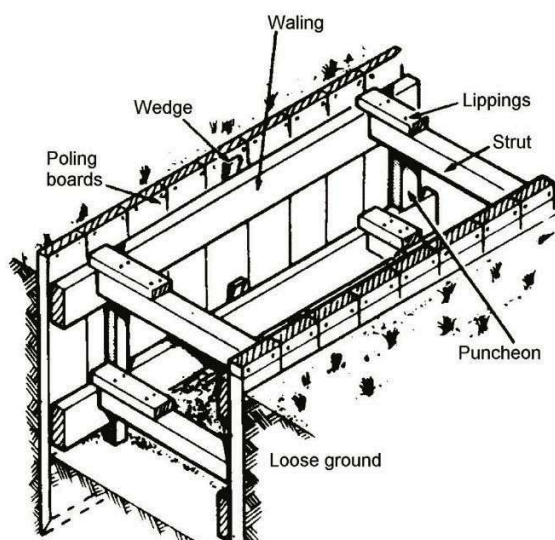
All erection, dismantling and alteration of any support must be under the supervision of a competent person.

6.3.3.1 Types of support systems

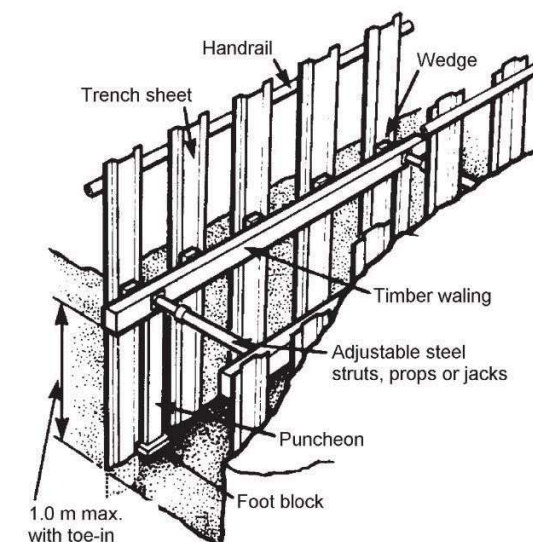
Timber shuttering, steel trench sheets and fixed or adjustable props may be used; the props can be mechanical or hydraulic.

A temporary framework of support of a protective box or cage type may be required to protect workers. This can be moved forward as timbering progresses.

The Construction Plant-hire Association has produced guidance on the management of shoring in excavations, hazard identification for risk assessment, as well as selection of proprietary shoring equipment. The guidance is aimed at anyone involved in the planning, management, design and supervision of excavation works.



Close-boarded excavation



Open sheeting using steel sheets and jacks or acro props

EXCAVATIONS

Proprietary support systems are available and include the following.

Shields (drag boxes), which are two vertical plates that are braced permanently apart; they are intended as temporary localised protection for workers, and can be dragged along by the excavator as work progresses.

Trench boxes, which are modular and are lowered in, and form a permanent support for the trench sides; they can be extended in height and width.

Plate lining systems have metal plates that slide into position between vertical soldier posts at preset intervals; the posts counteract stress from trench sides.

You need to be aware that any of these support systems can cause problems when underground services cross the line of the trench being excavated. The use of a hydraulic waling frame may be better used in this case, as spaces can be left between the metal sheets once they are in position. Hydraulic frames are ideal when excavating pits and inspection chambers, as they can be extended in both length and width.

6.3.4 Inspection and maintenance of support systems

Excavations must be closely monitored when they are first opened and sides are unsupported. The Construction (Design and Management) Regulations 2015 (CDM) state the following.

- All practicable steps must be taken to prevent danger to any person, including, where necessary, the provision of supports or battering, to ensure that:
 - no excavation or part of an excavation collapses
 - no material forming the walls or roof of, or adjacent to, any excavation is dislodged or falls
 - no person is buried or trapped in an excavation by material which is dislodged or falls.
- Suitable and sufficient steps must be taken to prevent any person, work equipment or any accumulation of material from falling into any excavation.
- Suitable and sufficient steps must be taken, where necessary, to prevent any part of an excavation or ground adjacent to it from being overloaded by work equipment or material. All excavations must be inspected by a competent person at the start of the shift, after any event that may have affected the strength or stability of the excavation, or if material unintentionally falls or is dislodged. No work should be undertaken until the inspection has taken place and the excavation is safe (*for further information refer to 6.5 Inspections and reports*).

Small movements of 6–12 mm are usually the only sign of weakening in cohesive soils – be aware of them, as they can pass unnoticed but they mean something is wrong. Movement can be seen from slight distortion in the timbering, bowing of poling boards and walings, or signs of local crushing. The ground can also dry out and shrink, which loosens timbering. Timber can be displaced by the ground absorbing additional moisture.

- Avoid damage to struts or walings when moving loads into or out of the excavation.
- Check timber regularly – it can dry out, shrink or rot.
- In bad weather spoil heaps can slump and loose boulders or masonry can fall into the excavation.
- A guideline is that the distance from the edge of the trench and the bottom of the spoil heap must be equal to or greater than the depth of the excavation. For example, the bottom of a spoil heap should be a minimum of 1 m from the edge of a trench that is 1 m deep.
- Do not allow heavy vehicles near the edge of an excavation unless the support work has been designed to permit it.
- Take care not to affect the stability of any neighbouring structure or features.
- Take into account the Party Wall etc. Act 1996 and any surveys, permissions or restrictions.
- Take precautions to protect workers (and others) before and during excavation works.



An unsupported excavation

In the example pictured here to the right, this excavation was only about 650 mm deep, and it was left unsupported overnight.

There was heavy rain during the evening, and the team returned the following morning to find this collapse, which had put the existing cast-iron gas main at risk.



6.4 Excavation safety

6.4.1 Access and egress

There must be adequate and safe arrangements in place for getting into (access) and getting out of (egress) an excavation.

This is usually achieved with ladders, which need to be properly located and fixed with suitable overruns or other handholds. Adjustable, lightweight staircases or towers are becoming increasingly common, and are safer and quicker to use.

Under no circumstances should anyone use the side supports or underground services that cross or run along the trench as footholds to climb into or out of an excavation.

The risk assessment should also consider emergency escape and recovery of casualties. Ideally, provision should be made to have a hoist installed that is equipped with the capability to lift a stretcher from the base of the excavation.

First aid needs, provision of trained personnel and suitable equipment for the hazards that may be present must also be accounted for.



Safe access into a deep excavation, with all-round secure edge protection



Climbing into or out of an excavation using walings, underground services or struts must be prohibited. This must be specifically covered in the safe system of work and/or method statement.

6.4.2 Guarding excavations

Suitable steps must be taken to prevent any person, vehicle, plant or equipment, or any accumulation of earth or other materials, from falling into an excavation. These include the following

- Robust physical barriers and toe boards. Barriers must be capable of resisting impact or the weight of a person leaning on them. They should be used to keep people safe and materials, plant and equipment away from the edges. (Pin-and-rope or rolls of orange plastic mesh are not suitable.)
- When barriers are removed to facilitate work, they should be replaced as soon as possible.
- The edges of an excavation should be lit during darkness, particularly if close to public roads and spaces.
- If excavation work is carried out on the highway, Local Authority approval is required and appropriate signing and guarding complying with the *Traffic Signs Manual*, chapter 8, and *Safety at Street Works: A Code of Practice* must be utilised.
- Ensure adequate signs and hazard warning lights are used when it is dark or during foggy conditions.



Excavation edge protection barriers



If plastic chapter 8-type barriers are used, such as during street works and road works on the public highway, they must be installed to be robust enough to stop a person who falls or leans against them from falling into the excavation.

This is particularly important if the barriers are being used to protect the public.



For further information on guarding excavations refer to Chapter F01 Street works and road works.

A series of images presented on the next page demonstrate a number of safe excavation systems in practice, with features shown including edge protection, guarding, and access and egress.

e.g. Safe systems in practice



Deep inspection chamber excavation with good access and edge protection



Deep inspection chamber excavation with good access, edge protection and additional ladder



Protected excavation



Drag boxes with guard-rail attachments and ladder access is good practice. Plastic orange netting, which is not suitable guarding for this type of excavation work, is bad practice



Remote controlled trench compactor avoiding the need to enter the excavation



Designated bucket changing area



Fencing along utility trench

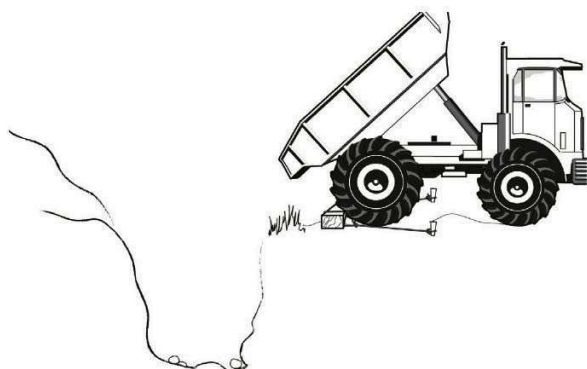
6.4.3 Vehicles and plant

Plan traffic routes so that no vehicles or plant go near excavations, other than those that need to complete the task at hand.

Be aware of surcharging when heavy items of construction plant are near to an excavation. Where this cannot be avoided, it must be considered in the design of the support system that will be installed.

Stop blocks must be used if vehicles are required to tip into an excavation. Ensure these are placed far enough away to avoid damage to the edge of the excavation.

Use a vehicle marshaller to give directions to the vehicle driver or plant operator.



Stop blocks should be used, where appropriate



Safe segregation of excavation work and plant



During 2016, a large number of incidents involving forward tipping dumpers resulted in many serious injuries and six fatalities in six months. These mainly involved the dumpers overturning and collisions with pedestrians. Many of the fatalities from overturns were a result of the operator not wearing the seat belt and attempting to jump clear, but becoming caught and crushed by the overturning machine.



For further guidance refer to the HSE publications *The safe use of vehicles on construction sites* (HSG144) and *Construction site transport safety: Safe use of site dumpers* (CIS52) which can be downloaded free from the HSE website.



For further information refer to Chapter C06 Mobile work equipment.

6.4.4 Site lighting

There must be suitable and sufficient lighting at every workplace, the approaches to it and on traffic routes. During hours of darkness or other periods of reduced natural light, excavations and the guarding surrounding them will need to be illuminated, particularly if the public have access to those areas.

You must pay particular attention to ensuring there is adequate lighting for access points, openings and lifting operations. The workforce should be wearing suitable high-visibility clothing. It is important to clean and maintain high-visibility clothing so that it remains effective in low light or during the hours of darkness.

6.4.5 Ventilation

Excavations must be kept clear of suffocating, toxic or explosive gases. Gases can seep through the soil and accumulate in the bottom of the excavation.

Naturally-occurring gases include hydrogen sulphide, as one example: a colourless, toxic gas that can be created by the decomposition of organic matter such as rotting vegetation or sewage, which if inhaled can cause convulsions and cardiac arrhythmias.

Gases that are heavier than air (such as plant exhaust emissions) can leak or roll into an excavation and displace the air, which can lead to asphyxiation.

Leakage of propane and butane from liquefied petroleum gas (LPG) cylinders is potentially dangerous; any gas leak will sink to the lowest point and form an explosive concentration that cannot disperse naturally.

The excavation will need to be checked and continuously monitored for the presence of gases and purged before entry by workers if these are found.



For further information on ventilation refer to Chapter B07 Control of substances hazardous to health.

6.4.6 Confined spaces

An excavation could be designated as a confined space due to any of the following hazards being present.

- Risk of engulfment, drowning or entrapment by a free-flowing solid (such as sand and fine soils) or a sudden rush of groundwater or rainwater.
- Nature of the excavation (deep and narrow) and the ground in which it is dug, which could result in the accumulation of a naturally occurring gas, such as hydrogen sulphide (H_2S).

EXCAVATIONS

- Risk of asphyxia due to the nature of the ground. Peaty soils absorb oxygen from the atmosphere so the bottom of a trench dug in peat is unlikely to have a sufficient level of oxygen. The action of chemicals in the soil on brownfield sites can also alter the oxygen content in an excavation. Soil which is high in limestone can produce carbon dioxide if it is contaminated with acidic rainwater.
- Substances used (such as LPG), which are heavier than air.
- Presence of carbon monoxide (CO) from internal combustion engines within the excavation. Though slightly lighter than air, CO may build up in the excavation and take time to dissipate (clear).
- Nature of the substances released from any underground services that have to be broken into.
- Extremes of temperature leading to hypothermia or heat exhaustion.
- Dust and fumes from mechanical cutting equipment or welding.

All of these factors can potentially reduce the percentage of oxygen in the air and/or introduce toxic, flammable or explosive gases into the air. Control measures on the risk assessment may include forced fresh-air ventilation or, as a last resort, appropriate respiratory protective equipment.



For further information refer to Chapter D08 Confined spaces.

If an excavation is classified as a confined space, tests for toxic gas or oxygen depletion must be carried out before work starts and continuously as work progresses.

It is also recommended that the work should be subject to the issue of a permit to work certificate, and that the first consideration should always be what measures can be taken to enable the work to be carried out without the need to enter the confined space.

6.4.7 Underground cables and services

Before any excavation works take place, you must ensure underground services (such as gas, electricity, water and telecommunications) have been identified. Many injuries each year are caused by accidental contact with underground services. You must have controls in place to reduce the risk of injury from underground cables or other underground services.



For further information refer to Chapter D07 Underground and overhead services.

6.5 Inspections and reports

An inspection of an excavation must be carried out by a competent person at the following intervals.

- At the start of each shift during which work is to be carried out.
- After an event that is likely to have affected the stability of the structure.
- After the accidental fall or dislodgement of any material.

The competent person must be satisfied that work can be carried out safely and a report, containing the following information, must be prepared.

- Name and address of the person on whose behalf the inspection was carried out.
- Location of the work being inspected.
- Description of the place of work, or part of that place of work inspection, including plant and equipment or materials.
- Date and time of the inspection.
- Details of any issues identified that could give risk to the health and safety of any person.
- Details of any action taken as a result of any issues identified.
- Details of any further actions considered necessary.
- Name and position of the person making the report.

If, following inspection, the competent person is not satisfied that work can be carried out safely then no work should be carried out until the matter has been satisfactorily remedied. All reports must be completed and written by the competent person, and a copy provided within 24 hours to the person on whose behalf it was undertaken.

A copy of the report must be kept on site until the work is completed and for three months, generally at the company head office, from the date of completion. You do not need to do more than one written report in any period of seven days. However, it is advisable that a daily record of inspection is kept (for example, in the site diary). There is no statutory form for inspections and information can be kept electronically. However, you must be able to produce a hard copy for an HSE inspector, if so requested.

6.6 Excavators used as cranes

Excavators are classed as mobile work equipment and are therefore subject to the requirements of the Provision and Use of Work Equipment Regulations 1998 and, when being used for lifting operations, the Lifting Operations and Lifting Equipment Regulations 1998.

Excavators and backhoes are principally designed for earth-moving operations and not lifting operations as a main function. The planning for a lifting operation using excavators should primarily consider whether these machines are suitable for the required lifting activity. As a lifting operation, it is subject to suitable and sufficient planning by a competent person, including a risk assessment and a safe system of work, that takes into account the capabilities of the excavator for the selected task.

The risk assessment and method statement should take into account that when an excavator is in the object handling mode (being used as a crane), it will be necessary for the slinger to approach the machine to attach or detach the lifting accessories. They will then be within a hazardous area (the slewing arc of the boom and dipper) and be at risk of being struck by the load, bucket or excavator arm.

Excavator operators and slingers must be made aware of these dangers; effective communication and constant vigilance are essential.



When the slinger is within the slewing arc of the machine's boom to attach or remove lifting accessories, the operating controls of the machine must be isolated.

The weight of the bucket (if still fitted), plus the quick hitch coupler must be added to the weight of the load to establish if the machine will be working within its safe working load (or rated capacity).

The bucket should be removed to improve the excavator operator's visibility. If a quick hitch coupler is fitted, the operator must ensure that the link is facing downwards (operating ram extended) so that any part of the lifting accessory will not foul the coupler, allowing the accessory to hang free at all times.

The risk assessment must also address the following.

- The need for the lifting operation to be ideally segregated from other work activities taking place in the vicinity, particularly where it is necessary for the machine to travel with a suspended load.
- Ground conditions, particularly where an excavator will carry out the lifting operation, which should be on level, firm terrain. Where a pick and carry operation is required, the intended travel route must be checked (for both tracked and wheeled types) so that there are no voids, it is clear of trenches, is on firm, level ground, has no obstructions and personnel are safely segregated from the travel route.

A rated object handling capacity table must be available in the cab. If the rated lifting capacity for an excavator (or the backhoe of a backhoe loader) is greater than 1 tonne (or the overturning moment is greater than 40,000 Nm), the machine must be fitted with the following.

- A boom lowering control device on the raising boom cylinder(s) (a safety check valve), which meets the requirements of BS ISO 8643 (*Earth-moving machinery. Hydraulic excavator and backhoe loader lowering control device. Requirements and tests*).
- An acoustic or visual warning device, which indicates to the operator when the object handling capacity or corresponding load movement is reached.

Where the upper structure of a 360° excavator needs to slew, the machine's lifting capacity is reduced when slewing away from the front or rear of the tracks or axles. This is because the track or wheel width is less than the track length or wheel base.

Chains or slings for lifting must not be placed around or on the teeth of the bucket. Accessories for lifting must only be attached to a purpose-made point on the machine.

Lifting points are manufactured with the machine. Unofficial ad-hoc additions (such as after-market hooks or points welded onto machines) are highly dangerous and not acceptable.

Where the risk assessment shows that there is a significant risk of overloading or overturning of the excavator, the fitment of a rated capacity indicator may be required.



For further guidance refer to the CPA document *Guidance on lifting operations with 180° and 360° excavators*.



For further detailed guidance refer to Chapter C08 Lifting equipment.

EXCAVATIONS – APPENDIX A

Appendix A – Safety awareness around excavators

This information covers basic safety rules that should be followed by the operator, vehicle plant marshaller (VPM) and others working around both tracked and wheeled excavators to reduce the risk of injury and allow everyone to go home safely. Plant operating and plant movements are a significant risk for everyone on site, not just those working directly around the excavator.



Accidents involving moving plant result in many serious injuries and fatalities every year in the construction industry.

If work is not managed safely people working around excavators are at high risk of sustaining serious life changing injuries or death. In confined work areas or where more than one piece of plant is in use, near misses, collisions and plant damage are also highly likely.

Technical advances in machinery (such as a remote operated cut-off switch) and familiarisation periods for new machines coming on to site help to reduce incidents and accidents.

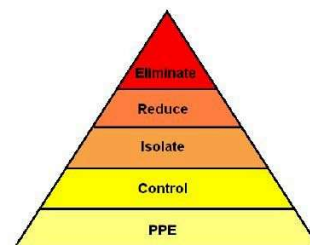
Communication and risk assessment



Before beginning any activity, a pre-task briefing should be carried out to ensure everyone involved is familiar with the task, and is aware of any risks and the control measures required. Any interface between different tasks should also be discussed to co-ordinate activities.

A risk assessment of the task should reflect the hierarchy of risk control to eliminate, reduce, isolate, control and, as a last resort, provide personal protective equipment.

- Does the workforce need to be working around the excavator?
- Where personnel are close to plant, are exclusion zones in place and are the required safety critical devices fitted to the excavator fully functional?
- Are vehicle plant marshals (VPM) required for the safe system of work?
- Is everyone wearing appropriate high-visibility clothing, which is in good order?



Safety zones



When working around an excavator it is important to be aware of the operator's limited all-round vision and the safety zones for the machine.

Where a risk assessment has identified the need for a competent VPM to manage the excavator movements then all instructions by the VPM must be followed. A VPM is responsible for managing the movement of people and vehicles safely around the excavator.

A vehicle marshaller carries out a separate role and supervises the general movement of vehicles on site (for example, deliveries).

Before entering the radius of the excavator, wait in an area of safety and acknowledge the machine operator or, where provided, VPM. The VPM will then instruct the excavator to stop, power down the engine, apply the hydraulic isolation lever and turn the engine off. Once the operator has acknowledged you it is then safe to approach.

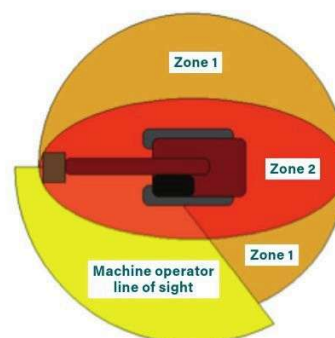


Approach the excavator cabin in the operator's clear line of sight.

You should only approach the excavator cabin in the operator's clear line of sight. Approaching the excavator from any other angle should be avoided (Zone 1 in the illustration). Where this is not possible (such as when an excavator is working near to a pedestrian route) the machine operator or VPM must acknowledge your need to approach and the correct shut-off procedure must be followed, as described above, before personnel enter the exclusion zone.

The machine operator should only restart the machine once they have seen you leave the exclusion zone or the VPM has confirmed that it is safe to do so.

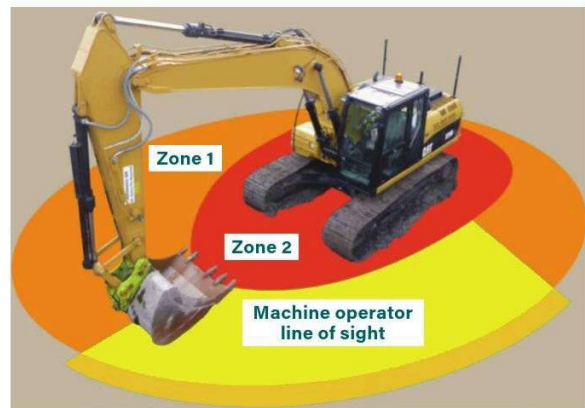
Pedestrians often still approach an excavator from the operator's blind side, assuming the operator is unlikely to move or that they can be seen by the operator. It is important that all personnel on site are aware of the restricted visibility in excavators. An appreciation of this can be achieved by allowing non-excavator operators to sit in the cab in controlled conditions to experience the restrictions for themselves. Stickers on the machine can be used as a reminder.



e.g. Danger! You are approaching on my blind side. I cannot see you.

It should not be necessary to enter the area directly around the machine or underneath the boom (Zone 2 in the illustration). Any tasks where this is unavoidable should be individually considered and have a site and task specific safe system of work put in place. These activities may include the following.

- Drainage or pipe laying.
- Archaeology.
- Piling.
- Earth moving.
- Bearing tests.
- Lifting operations.
- Plant maintenance.



Always gain permission from the vehicle plant marshal before entering an excavator's working area.



Safety zones

Using the safety zones guidance, spot what is wrong in this example.

- Supervisor and workforce are on the blind side of the plant operator.
- No exclusion zone is visible if operatives are to be excluded from the works area.
- No visible control of excavator. No eye contact with operator, and no VPM present.
- No-one is taking the decision to work safely or not at all.



Appendix B – Case study – Excavation collapses after dangers are ignored



Director and excavator operator jailed

A 52-year-old director of a house building company (working as the site manager at the time of the accident) was convicted of gross negligence manslaughter following the death of a worker on a construction project. The conviction followed a nine-week trial at Northampton Crown Court into the death of a 33-year-old father-of-five whilst he was employed as a ground worker at a building site in Collyweston, Northants.

The court heard evidence of how the worker had been standing next to a deep trench, which had been incorrectly excavated by the excavator driver, when a wall of the trench collapsed and buried him beneath the collapsed material. Despite the efforts of fellow workers he was pronounced dead at the scene after his body was recovered. Northamptonshire Police and HSE investigators found that the sides of the trench had not been properly or adequately secured and that the site manager (the company director) and excavator driver both ignored basic safety measures.

Northampton Crown Court convicted the defendants, as follows.

- The site manager and director of the company was found guilty of gross negligence manslaughter and received a four-year sentence. He will spend two years in custody and two years on license. He was also ordered to pay costs of £90,500.
- The self-employed excavator driver, who traded as a demolition contractor, was found guilty of a failure as a self-employed person to discharge his duty to ensure the health and safety of persons not in his employment, contrary to Section 3(1a) of the Health and Safety at Work etc. Act 1974. He was also found guilty of failing to take all practicable steps to prevent danger, contrary to Regulation 31 of CDM. He was given a 12-month sentence and will spend six months in custody and six months on license. He was ordered to pay £20,000 towards costs.

Speaking after the verdict a detective superintendent from Northamptonshire Police said: 'This was a tragic loss of a young life that could so easily have been avoided. The defendants were both experienced in ground works and failed to show even the most basic safety measures to prevent harm to workers – such as the young father who lost his life. He leaves behind five young children who are now being cared for by his parents. They were in court throughout the trial and have shown true dignity and strength despite listening to harrowing evidence. This has been a very lengthy and complex investigation and I would like to thank the jurors for their dedication and attention to detail during this long trial'.

06

Appendix C – Case study – Building contractor jailed after house collapse



Structural integrity of house undermined, leading to collapse

In April 2018 a building contractor was sentenced to seven months' imprisonment following an incident in February 2015 in which a house in Brighton partially collapsed.

Hove Crown Court heard how the builder undermined the structural integrity of the house by digging out the basement. He then failed to act on the advice of a structural engineer on how to remedy the situation, resulting in the gable wall partially collapsing and the ground floor collapsing into the basement. Adjacent properties had to be evacuated and the area cordoned off due to concerns that members of the public living nearby and passing through the area were at risk.

An investigation carried out by the HSE found that the property had bungaroosh walls, common to buildings built in the mid-18th to 19th century in the Brighton area. They are constructed with a mixture of rubble, timber, pebbles, stones, lumps of chalk, pieces of flint and anything else at hand, in a lime mix mortar set between shuttering. It has been described by some construction experts as 'the worst construction material in the world'. The porous composite is susceptible to rot and can become friable in either excessively damp or dry conditions.

The HSE said the builder did not have sufficient knowledge of bungaroosh and that those working on properties made from the legacy material 'must fully understand what they are dealing with' because the buildings are 'more challenging in terms of structural stability'.

The investigation also found that the builder failed to report the incident to the HSE as a dangerous occurrence, in accordance with the Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 2013 (RIDDOR).

The builder, of Woodingdean, Brighton, pleaded guilty to breaching Regulation 25(1) of CDM and Regulation 7 of RIDDOR. The defendant was sentenced to five months' imprisonment for count one and two months' imprisonment for count two, to run concurrently. He was also ordered to pay costs of £7,000.

The HSE principal inspector said: 'Basement work must be properly planned to ensure the structural integrity of the building throughout the construction work. When this type of work is done badly, workers and members of the public are at significant risk of serious injury or death. In addition, we cannot underestimate the impact on the homeowners when their properties are extensively damaged'.

(Source: HSE.)

CONTENTS

Underground and overhead services



7.1	Introduction	106
7.2	Important points	106
7.3	Underground services	107
7.4	Plan	108
7.5	Locate	109
7.6	Dig – excavating	110
7.7	Damage to underground services	112
7.8	Overhead services	113



GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

UNDERGROUND AND OVERHEAD SERVICES

Overview

This chapter provides practical guidance on factors to be considered when preparing to work with, or near to, underground and overhead services.

On the majority of occasions a licensed utility company will carry out this type of work. However, existing services, whether known or unknown, remain a risk on nearly all projects.

Always bear in mind that location drawings for services are not always accurate. Therefore, locations may not be correct and there may even be some services that have not been recorded.

This chapter also includes practical guidance for dealing with the hazards and the subsequent risks of working, or coming into contact, with overhead power lines.

7.1 Introduction

Underground services are, to a great extent, out of sight and out of mind until, perhaps, there is a fault or another reason to excavate. Every year people are injured and some killed due to accidental contact with underground services, such as electricity cables and gas pipes.

In every case, damage and injury could have been avoided if the proper procedures had been followed. The statistic of 70,000 instances of damage and 400 injuries per year as a result of coming into contact with overhead and underground electricity cables shows that there is considerable room for improvement.

Coming into contact with overhead services also includes the power lines serving the railway network.



For further information refer to Chapter F02 Trackside safety.

This chapter makes reference to plant or equipment (for example, cranes, excavators, mobile elevating work platforms (MEWPs) and scaffold poles) that may come into contact with overhead power lines.



The construction phase plan will provide information about hazards specific to a site, and must include locations and types of underground and overhead services that can be found in the area.

7.2 Important points

PLAN the work.

LOCATE the underground services.

DIG using a safe method of work.

7.2.1 Underground services

- Before doing anything yourself, check with the utility providers for existing services in the area that you are planning to work in.
- Ensure existing service routes are clearly marked on your plans.
- Use paint, tape or markers to identify existing services. Do not use metal spikes, which might penetrate a cable or pipe.
- Absence of marker posts is not evidence that there are no underground services.
- Damage to gas pipes can cause leaks, which can result in fire and/or explosion.
- The most common injury is from contact with electrical cables. These often carry high voltages and can result in death or major burns.



The exact position of a service will only be known once the service is uncovered.

7.2.2 Overhead services

- Where work cannot be avoided near to overhead services you must consult the local electricity supplier or distribution network operator (DNO) for advice before any work begins.
- A safe system of work must be planned and implemented for all work on or near to overhead electrical works.
- Ensure power lines are dead, or suitable precautions are taken.